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A Bayesian approach to generating distribution-based signals in pairs trading

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This paper introduces a novel approach to improving the precision and adaptability of trading signals in pairs trading. Our method derives the full conditional distribution of the hedge ratio and utilizes its quantiles as confirmation thresholds for trading signals generated within the standard cointegration framework. We apply this approach to 41 selected asset pairs across the U.S. and Brazilian markets, evaluating its effectiveness through empirical analysis. Our findings indicate that the proposed Bayesian hierarchical model improves trading performance and risk management in the majority of analyzed pairs, with particularly strong results in dual-class share configurations. The method achieves these improvements while reducing trading frequency by approximately 24%, which implies less exposure to transaction costs in practical implementations. By adopting a distribution-based framework, our approach not only enables more timely and adaptive trading signals but also enhances pair selection by effectively filtering out false positives in cointegration tests, as demonstrated through simulations.

Keywords: Statistical arbitrage; Pairs trading; Cointegration; Bayesian hierarchical models

JEL Classifications: C11, G11

1. Introduction

Pairs trading is a market-neutral investment strategy based on statistical arbitrage that exploits temporary mispricings between two assets that historically move together. When the price spread between the two assets diverges from equilibrium, it creates a potential arbitrage opportunity where investors aim to profit from the transitory divergence (Gatev *et al.* 2006). This strategy assumes that the spread will revert to its historical long-term mean due to the underlying economic forces holding the pair together (Avellaneda and Lee 2010). Pairs trading is attractive to investors because of its ability to generate low volatility returns that are largely uncorrelated with broader market movements. Further, the strategy can be implemented across various time horizons, with holding periods ranging from intraday trades lasting minutes to longer-term positions that may span several days, weeks, or even months. The combination of low volatility, minimal exposure to systemic market risk, and adaptability across asset

classes has made pairs trading a versatile tool in the portfolio of quantitative traders.

While the fundamental strategy behind pairs trading is conceptually simple, its practical implementation faces three key challenges that can significantly affect its effectiveness. First, accurately identifying pairs of stocks with a stable relationship over the desired timeframe can be difficult. Market conditions may disrupt assumptions of mean reversion, particularly during periods of stress when historical correlations break down. Traditional models often fail to account for non-Gaussian return distributions, fat tails, or volatility clustering, all of which are common in asset return data (Guillaume *et al.* 1997, Lux and Marchesi 2000, Cont 2001).

Second, allocating shares for the pairs strategy is problematic because the appropriate balance between long and short positions is the foundation of the self-financing strategy, which can be sensitive to price movements and market volatility (Alexander and Dimitriu 2002). In practice, fixed allocation strategies, such as equal dollar amounts or a simple ratio, may not account for time variations in market dynamics or the differing volatilities of the two assets. This can lead to disproportionate exposure to one asset, increasing the risk of loss if the asset moves unfavorably.

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The third challenge involves determining optimal entry and exit points for trades. Strategies such as the distance method or standard cointegration rely on historical thresholds or statistical deviations to trigger buy and sell signals. However, these signals may be unreliable when faced with market regime shifts or structural changes. Identifying the correct time to exit a trade, particularly when the spread fails to converge as expected, can result in significant losses if the market behaves unpredictably. Additionally, transaction costs and liquidity issues can further erode the strategy's profitability, making the precise timing of trades critical to the strategy's overall success.

In this study, we propose a novel approach that improves the pair selection process and optimizes entry and exit signals for pairs trading utilizing Bayesian hierarchical modeling. Our key contribution lies in adopting a truncated normal prior for hedge ratios (β_1) and using posterior quantiles derived from its conjugate distribution as a confirmation layer to enhance the precision of trading signals within the cointegration framework. In the proposed method, we can also improve pair selection by introducing a tolerance threshold for consecutive negative β_1 values in the Gibbs sampling process. By discarding pairs that exceed this threshold, the method ensures that only pairs exhibiting genuine co-movement are retained for trading.

Using Bayesian hierarchical modeling provides several advantages over the traditional pair selection and trading systems employed in the standard cointegration strategy. By incorporating a truncated normal prior and its full conditional conjugate for hedge ratios, we impose a structure that ensures hedge ratios remain within a realistic range, reducing the risk of extreme or unstable values. This contrasts with traditional approaches that rely solely on historical averages or static thresholds, which may not adapt well to changing market dynamics. Introducing a tolerance threshold for consecutive negative β_1 values serves as an additional safeguard, helping filter out pairs that are likely to break down in practice. This approach enables our method to be adaptive and resilient to regime shifts and market anomalies, ultimately improving trading performance.

Our empirical analysis demonstrates that the Bayesian framework delivers superior performance relative to standard cointegration methods across most analyzed pairs, achieving its strongest results when applied to dual-class share structures in emerging markets. The method's primary advantage lies in its capacity to produce trading signals that are both more accurate and contextually responsive to evolving market dynamics. Unlike conventional strategies that rely on rigid z-score thresholds, the Bayesian approach incorporates distributional information to generate signals that adapt dynamically to changing conditions. An important practical benefit emerges from the method's enhanced selectivity: by filtering low-confidence signals, it reduces trade execution frequency considerably, resulting in meaningful savings on transaction costs that would otherwise erode returns. Our simulation studies further validate the approach by demonstrating its effectiveness in identifying spurious cointegration relationships, a critical capability for avoiding unstable pair selections. This filtering mechanism proves particularly valuable in volatile or regime-shifting markets, where false

positives in cointegration tests can lead to substantial trading losses.

In addition, the ability of the Bayesian framework to incorporate uncertainty into the model further contributes to its improved performance. The method adapts to short-term fluctuations and long-term structural changes in asset behavior by accounting for variations in hedge ratios and market volatility. This flexibility enables the algorithm to better capture asset pairs' mean-reversion properties while minimizing exposure to false signals. Consequently, our results suggest that the proposed algorithm offers practitioners a more refined and sophisticated tool for pairs trading in comparison to standard cointegration, with enhanced adaptability and operational efficiency.

The remainder of this paper is organized as follows. Section 2 discusses the cointegration framework and its role in pairs trading, providing the theoretical background for the improvements introduced by our Bayesian framework. Section 3 develops the Bayesian hierarchical model and derives the full conditional distribution for hedge ratios, completing the methodological foundation of our approach. Section 4 describes the backtesting methodology employed in this study, whereas Section 5 presents the algorithm's performance compared to the traditional cointegration strategy in selected pairs from the US and Brazilian markets. Section 6 reports simulation studies focusing on the robustness of the method in identifying false positives in cointegration tests. Finally, Section 7 summarizes our findings and proposes avenues for future research.

2. Cointegration in pairs trading

In recent years, the literature on pairs trading has expanded considerably, with new methodologies aimed at enhancing the strategy's profitability and robustness. These advancements include the use of deep reinforcement learning techniques to create dynamic stop-loss boundaries on trades (Kim and Kim 2019, Brim 2020); clustering firm characteristics for more effective pair selection (Han *et al.* 2023); using an Ornstein-Uhlenbeck (OU) process to model prices and adopting a mean-variance (MV) framework to find the best portfolio strategies over time (Zhu *et al.* 2021); denoising stock price series using wavelet transforms (Eroğlu *et al.* 2023); applying state-space models to account for nonlinear and non-Gaussian features in asset prices (Zhang 2021); and developing advanced copula-based strategies to capture complex dependence structures and nonlinear relationships in pairs (Tadi and Witzany 2025).

Despite these innovations, cointegration-based methods remain widely popular due to their theoretical elegance and relatively straightforward application. The core of the cointegration strategy consists of selecting two stocks, say X and Y , that historically move together and whose individual price series (y_t and x_t) are non-stationary (Vidyamurthy 2004). Specifically, $y_t \sim I(1)$ and $x_t \sim I(1)$, where $I(1)$ denotes an integrated process of order 1, meaning that the series becomes stationary only after taking the first difference. After fitting a linear model relating Y to X , i.e. $\hat{y}_t = \hat{\beta}_0 + \hat{\beta}_1 x_t$, if the

residuals (e_t) are stationary, i.e. $e_t \sim I(0)$, where $I(0)$ denotes a stationary process, then the pair is said to be cointegrated.

The stationarity of the residuals ensures the mean reversion between Y and X , enabling predictable long-term behavior. Trading signals are generated from the standardized spread $z_t = (y_t - \hat{\beta}_1 x_t - \mu)/\sigma$, where μ and σ denote the mean and standard deviation of the spread $y_t - \hat{\beta}_1 x_t$ computed over the estimation window. The standardized spread is also stationary as a direct consequence of the residuals' stationarity (Vidyamurthy 2004). Positions are taken when $|z_t|$ exceeds a pre-specified entry threshold of k standard deviations, going long (or short) on $\hat{\beta}_1$ shares of stock X for each share of stock Y . Positions close either at mean reversion ($z_t = 0$) or at stop-loss thresholds ($|z_t| > k + \xi$), where $\xi > 0$ represents a pre-established loss margin measured in standard deviations of the spread beyond the entry level.

2.1. Testing for cointegration

The cointegration test was first introduced in the seminal paper by Engle and Granger (1987). The authors proposed using the Augmented Dickey-Fuller (ADF) unit root test to assess the presence or absence of stationarity in the original series and its residuals when regressing y_t on x_t . As previously mentioned, cointegration is present when two non-stationary series, after combined in a linear model, produce stationary residuals.

The original ADF procedure tests the null hypothesis that a time series has a unit root, implying that the series is non-stationary and follows a stochastic trend. A time series y_t is said to have a unit root if shocks to the series have a permanent effect, indicating that the series does not revert to a mean over time. The ADF test fits the following time series regression model:

$$\Delta y_t = \alpha + \theta t + \gamma y_{t-1} + \delta_1 \Delta y_{t-1} + \delta_2 \Delta y_{t-2} + \dots + \delta_p \Delta y_{t-p} + \epsilon_t$$

where $\Delta y_t = y_t - y_{t-1}$ represents the first difference of the time series, and $\Delta y_{t-m+1} = y_{t-m+1} - y_{t-m}$, $m = 2, \dots, p+1$, denotes the m th difference between consecutive observations in the series. The θ term represents the deterministic trend, and the δ_i terms capture the auto-regressive dynamics of the first differences of the series. In this model, γ is the key parameter for testing the presence of a unit root. If $\gamma = 0$, then y_t has a unit root, i.e. the series is non-stationary and follows a random walk. In contrast, if $\gamma < 0$, the series is stationary and reverts to a long-term mean.

Subsequent methodologies were proposed as more robust alternatives to the original Engle-Granger framework. Among several methods (Johansen 1988, Toda and Yamamoto 1995, Gregory and Hansen 1996, Banerjee *et al.* 1998, Pedroni 1999, Saikkonen and Lütkepohl 2000, Pesaran *et al.* 2001, Pedroni 2004, Westerlund 2007, Bracegirdle and Barber 2012), the Phillips-Perron (PhP) test (Phillips and Perron 1988) is one of the most commonly adopted alternatives, having the notable advantage of providing a nonparametric approach to handling nuisance parameters. The test consists of a variation of the Augmented Dickey-Fuller procedure that takes a different

approach to handling autocorrelation and heteroscedasticity when testing for the presence of a unit root in the series.

In the ADF test, lagged difference terms (Δy_{t-k}) are included in the regression to correct for autocorrelation in the residuals. These lags help ensure that the regression residuals are approximately white noise (i.e. uncorrelated and homoscedastic). However, the effectiveness of the ADF test depends on the proper selection of the number of lagged terms, which may require additional model selection criteria. Even with this correction, there are cases where the residuals remain autocorrelated or heteroscedastic, violating the assumption of white noise and potentially leading to biased results.

In contrast, the Phillips-Perron test does not include lagged terms in the regression model. Instead, it directly adjusts the test statistic using heteroscedasticity and autocorrelation consistent (HAC) standard errors. The method builds on the foundational work of Huber (1967) and White (1980), who developed techniques for robust covariance estimation to handle heteroscedasticity. Newey and West (1987) later extended these ideas to create a unified approach that addresses both heteroscedasticity and autocorrelation.

The regression model used in the PhP test mirrors the basic Dickey-Fuller framework but omits the lagged terms typically included to address autocorrelation:

$$\Delta y_t = \alpha + \theta t + \gamma y_{t-1} + \epsilon_t,$$

where $\Delta y_t = y_t - y_{t-1}$ represents the first difference of the series, and ϵ_t is the error term, which may exhibit autocorrelation or heteroscedasticity. The PhP procedure modifies the test statistic associated with γ (the parameter used to test for the presence of a unit root) by applying nonparametric corrections to account for these issues. This approach eliminates the need to explicitly model the residual structure using lagged terms, making the test less sensitive to user-defined parameters.

By applying robust variance and covariance estimators, the Phillips-Perron test also accommodates a broader range of time series models, including those with heterogeneously distributed innovations. These characteristics make the Phillips-Perron test a more flexible and reliable tool for practical applications, particularly for economic and financial time series data. However, like any other hypothesis test, the PhP test is also susceptible to the occurrence of false positives. The Bayesian approach we introduce in Section 3 significantly enhances the pair selection robustness by identifying a considerable proportion of false positives as demonstrated in Section 6.

3. Bayesian approach to distribution-based trading signals

3.1. Introduction

In this section, we introduce our new Bayesian approach to pairs trading, which builds upon the cointegration framework by dynamically enhancing the reliability of pair selection and refining the trading signals through probabilistic evaluation of the hedge ratio β_1 . We contend that β_1 contains critical

information about the pair relationship, such that any breakdown in the ‘co-moving’ assumption will likely manifest as changes in the pair’s hedge ratio. By quantifying the deviations of the current β_1 from its expected distribution, we implement an additional confirmatory layer for entry and exit signals generated by the standard cointegration strategy.

The core of the strategy lies in generating the full conditional distribution of the parameter vector, $\beta = [\beta_0 \ \beta_1]^\top$, through a hierarchical Bayesian model. Here, β_0 represents the intercept, and β_1 is the hedge ratio defining the relationship between the cointegrated assets. An additional benefit of our approach is that sampling from the full-conditional distribution of β_1 serves as an enhanced pair selection filter, significantly reducing false positives in the cointegration test—a result we demonstrate empirically in Section 5 and more comprehensively through simulation studies in Section 6.

With the full conditional distribution of β_1 derived, we identify two quantiles (e.g. 5th and 95th) from this distribution and utilize them as confirmatory triggers for placing both entry and exit orders. If the z-score of the spread $y_t - \hat{\beta}_1 x_t$ at decision period t crosses[†] the predefined boundaries (e.g. $\pm 2\sigma$, where σ represents the standard deviation of the spread), and the current $\hat{\beta}_1$ is well-behaved,[‡] i.e. within the chosen quantiles of the hedge ratio’s distribution, the entry signal is confirmed and the orders are executed. Otherwise, the entry signal is canceled.

Stop-loss thresholds are also established using standard deviations of the standardized spread, e.g. $\pm|2\sigma + \xi|$, where ξ is a pre-established loss margin measured in standard deviations. Once an operation is opened, if the standardized spread crosses the defined stop-loss boundaries in a future period, the algorithm emits a stop-loss signal that is only confirmed if the updated $\hat{\beta}_1$ is outside the defined quantiles of the hedge ratio’s distribution. Otherwise, the exit signal is canceled and the operation continues.

In our method, the take-profit threshold is set using a pre-specified margin of gain based on the standardized spread and will not depend on the current value of the hedge ratio. The common choice is $z = 0$, given that it represents the long-term equilibrium of the stationary process.

We hypothesize that the proposed method is profit-wise superior to the regular cointegration framework because it may

- (i) Enhance the pair selection step;
- (ii) Generate more timely trading signals.

If a pair is cointegrated and the two shares truly move together, it is reasonable to assume that the slope of the linear model $y_t = \beta_0 + \beta_1 x_t + \epsilon_t$ is positive. Based on this assumption, we assign a normal distribution truncated at $\beta_1 > 0$ as the prior for β to reflect the expected behavior of the pair. The full conditional will also follow a truncated normal distribution, as demonstrated in Section 3.2. If, when drawing realizations from the full conditional of β , we obtain a sequence

of m (e.g. $m = 30$) negative β_1 values, we opt to discard the pair due to the strong contradiction between the sampled values and our prior belief. This approach is primarily heuristic, grounded in the observation that cointegrated pairs that move together will have positive hedge ratios ($\beta_1 > 0$). While this is not a strict theoretical result, it aligns with the practical behavior of pairs trading strategies. In this way, our method can perform pair selection by confirming or discarding the results of the cointegration test, adopting this heuristic to guide decision-making.

Another essential component of our strategy is improving the precision of trading signals. Relying exclusively on the standardized spread for signal generation can result in entering trades just as the pair is about to lose its long-term equilibrium (cointegration) status or when the pair balance given by β_1 is excessively distorted. These two scenarios can lead to losses because the pair will not return to the long-term mean or because of an unbalanced share allocation. Furthermore, it is not uncommon for pairs trading practitioners to observe early stoppages that considerably affect the strategy’s performance. We hypothesize that the adoption of the distribution of β as an extra layer of confirmation contributes to generating more trustworthy and timely trading signals, thus mitigating the aforementioned risks and improving the strategy’s profitability.

We empirically evaluate the performance of our method against the standard cointegration strategy in terms of profitability and risk metrics in Section 5. The patterns of cumulative return factors and drawdowns provide insights into the differences in trading signals between the two methods, aiding in the assessment of hypothesis (ii). In Section 6, we further investigate hypothesis (i) by applying the proposed method to simulated cointegrated and non-cointegrated time series. This allows us to evaluate whether the Bayesian approach can correctly discard false positive indications from the cointegration test.

3.2. Theoretical background

The Bayesian representation of the regression model $y = X\beta + \epsilon$ consists of a hierarchical model comprised of prior distributions for the model parameters and a likelihood function representing our data model. When combined, these distributions allow us to compute the full conditional distribution for β , which is the distribution of interest in our proposed method of generating dynamic trading signals.

We assume that y follows a multivariate normal distribution

$$y \sim \mathcal{N}(\mu = X\beta, \Sigma = \sigma^2 I)$$

with $\beta = [\beta_0 \ \beta_1]^\top$, and analytical form for the likelihood given by

$$p(y | \beta, \sigma^2) = |2\pi \Sigma|^{-1/2} \times \exp\left(-\frac{1}{2}(y - X\beta)^\top \Sigma^{-1}(y - X\beta)\right). \quad (1)$$

Since our approach focuses on cointegrated pairs, the residuals are expected to be stationary by construction. This

[†] Assuming a discrete time span in which $t + 1$ is the trading time, $\hat{\beta}_1$ is estimated using data from the start of the analyzed period until $t - 1$.

[‡] The current $\hat{\beta}_1$ is estimated using data from the start of the analyzed period until decision time t , just before the trading time $t + 1$.

property justifies the assumption of homoscedasticity and uncorrelated errors ($\Sigma = \sigma^2 I$) within the Bayesian framework. Furthermore, it simplifies computation and aligns with our goal of dynamically updating posterior distributions for β without imposing unnecessary complexity.

As for our priors, we propose an Inverse Gamma distribution for σ^2 to model its strictly positive domain and accommodate skewness

$$\sigma^2 \sim \mathcal{IG}(q, r),$$

where $q > 0$ is the shape parameter and r^{-1} is the scale parameter with $r > 0$. The analytical form for its probability density function (pdf) is

$$p(\sigma^2 | q, r) = \frac{1}{r^q \Gamma(q)} (\sigma^2)^{-(q+1)} \exp\left(-\frac{1}{r\sigma^2}\right). \quad (2)$$

For β , we adopt a truncated multivariate normal prior to enforce positivity on its component β_1 :

$$\beta \sim \mathcal{TN}(\mu_\beta, \Sigma_\beta, 0, +\infty).$$

The pdf of this truncated normal distribution is proportional to a standard multivariate normal pdf, but with support restricted to $\beta_1 > 0$:

$$\begin{aligned} p(\beta | \mu_\beta, \Sigma_\beta, \beta_1 > 0) \\ \propto |\Sigma_\beta|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(\beta - \mu_\beta)^\top \Sigma_\beta^{-1}(\beta - \mu_\beta)\right) \mathbf{1}_{\beta_1 > 0}, \end{aligned} \quad (3)$$

where $\beta_1 > 0$ ensures the positivity constraint on the second element of the β vector.

We initialize μ_β and Σ_β using the sample regression estimator and the sample variance-covariance matrix (Rencher and Schaale 2008) based on all data collected up to time $[t-1]$. Specifically, we set $\mu_\beta = (X^\top X)^{-1} X^\top y$ and $\Sigma_\beta = \hat{\sigma}_0^2 (X^\top X)^{-1}$, where $\hat{\sigma}_0^2 = \frac{1}{n-2} y^\top (I - H) y$ and $H = X(X^\top X)^{-1} X^\top$. These choices follow standard Bayesian linear regression practices, using ordinary least squares (OLS) estimates as informative hyper-parameters. In addition, based on empirical evidence from our backtesting of the pairs trading strategy, we fix the variance Σ_β at $\hat{\sigma}_0^2 (X^\top X)^{-1}$. This approach enhances model stability and identifiability by reducing the uncertainty associated with β , a technique that has proven effective in similar contexts (see, e.g. Aguilar and West 2000, Chib *et al.* 2006 for applications in factor models). For further details on initialization methods in Bayesian regression, see Hooten and Hefley (2019).

3.2.1. Full conditional distributions. The choice of a truncated normal distribution for β is justified by the co-moving assets assumption and the adoption of the cointegration framework as the foundation of our pairs trading strategy. If two assets are predominantly moving in cooperation and exhibit cointegration, it is expected that their hedge ratio β_1 will remain positive throughout the period analyzed. This is aligned with Gelfand *et al.* (1992) suggestion, where the

authors indicate the truncated normal as the prior distribution in cases where the parameter of interest cannot be negative.

However, generating distributions directly from a truncated normal distribution can be computationally challenging. Following the recommendation by Gelfand *et al.* (1992), we instead derive the unconstrained full conditional distribution and impose truncation through rejection sampling.

The full conditional distribution for β will thus have the form

$$p(\beta | y, \sigma^2) \propto \begin{cases} p(y | \beta, \sigma^2) p(\beta), & \beta_1 > 0 \\ 0, & \text{otherwise} \end{cases}$$

The analytical form of the unconstrained full conditional will be

$$p(\beta | y, \sigma^2) \propto \exp\left(-\frac{1}{2}(-2b^\top \beta + \beta^\top A \beta)\right),$$

where $A = (X^\top (\sigma^2 I)^{-1} X + \Sigma_\beta^{-1})$ and $b = (X^\top (\sigma^2 I)^{-1} y + \Sigma_\beta^{-1} \mu_\beta)$.

After using further manipulation (see Hooten and Hefley 2019), we observe that the full conditional distribution of β has the kernel of a Normal distribution with mean $A^{-1}b$ and variance A^{-1} .

$$\beta | y, \sigma^2 \equiv \mathcal{N}(\mu_{\beta|y,\sigma^2} = A^{-1}b, \Sigma_{\beta|y,\sigma^2} = A^{-1}).$$

According to Gelfand *et al.* (1992), once we know the form of the unconstrained full conditional for the parameter of interest, we can then draw realizations from this distribution using a Gibbs sampler. This procedure is straightforward and can be done by simply generating from the unconstrained full conditional and retaining the variate value only if it falls in the cross-section constraint region.

The conjugate full conditional distribution for β truncated at $\beta_1 < 0$ is then

$$\begin{aligned} \beta | y, \sigma^2, \beta_1 > 0 \\ \equiv \mathcal{TN}(\mu_{\beta|y,\sigma^2} = A^{-1}b, \Sigma_{\beta|y,\sigma^2} = A^{-1}, 0, +\infty), \end{aligned}$$

with the analytical form given by:

$$p(\beta | y, \sigma^2, \beta_1 > 0) \propto \exp\left(-\frac{1}{2}(-2b^\top \beta + \beta^\top A \beta)\right) \mathbf{1}_{\beta_1 > 0}. \quad (4)$$

It is important to note that the full conditional distribution for σ^2 will also be affected by the constraints imposed to β . Once we have the form of the unconstrained full conditional, we can use the same procedure as before, retaining the variate value only if it falls in the cross-section constraint region.

The analytical form of the constrained full conditional for σ^2 is

$$p(\sigma^2 | \mathbf{y}, \boldsymbol{\beta}, \beta_1 > 0) \propto (\sigma^2)^{-(q+\frac{n}{2}+1)} \exp \left(-\frac{1}{\sigma^2} \left(\frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \frac{1}{r} \right) \right) \mathbf{1}_{\beta_1 > 0}. \quad (5)$$

The result above clearly depicts the kernel of an Inverse Gamma distribution. Note that q is the shape parameter whereas r^{-1} is the scale parameter of the prior Inverse Gamma distribution.[†] By letting $\tilde{q} = q + \frac{n}{2}$ and $\tilde{r} = [\frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \frac{1}{r}]^{-1}$, we have the unconstrained full conditional distribution for σ^2 as a conjugate Inverse Gamma (Hooten and Hefley 2019)

$$\sigma^2 | \mathbf{y}, \boldsymbol{\beta} \equiv \mathcal{IG}(\tilde{q}, \tilde{r}).$$

4. Backtesting methodology

This section presents the backtesting methodology used to compare the performance of the proposed Bayesian approach against the traditional cointegration method (referred to as the ‘reference’ strategy). We evaluate both strategies using standard performance metrics: cumulative return factor (CRF), volatility, maximum drawdown, Sharpe ratio (Sharpe 1966), and Sortino ratio (Sortino and van der Meer 1991). The CRF is defined as the ratio of terminal wealth to initial capital, i.e. $\text{CRF} = K_T/K_0$, where K_0 represents initial capital and K_T denotes terminal wealth. This multiplicative metric provides an intuitive interpretation of strategy performance, with $\text{CRF} > 1$ indicating profitable outcomes, while $\text{CRF} < 1$ meaning capital erosion. We adopt this representation rather than percentage returns because it naturally accommodates compounding effects and facilitates direct comparison across strategies with different holding periods.

4.1. Data

Our data comprises pairs from both U.S. and Brazilian markets. We obtained intraday adjusted close price data at 30-min and 5-min frequencies from two primary sources: Alpha Vantage Inc. for U.S. markets and MetaTrader 5 (MT5) database via XP Investimentos for Brazilian markets. The U.S. dataset spans over ten years (February 1, 2015, to January 31, 2025), while Brazilian data covers approximately five years (June 22, 2020 to January 31, 2025) due to MT5 API limitations. When data availability constraints affected either asset in a pair, we restricted analysis to overlapping periods to ensure consistency. The 30-min timeframe guided signal identification, while the 5-min data enabled precise price execution, creating a more realistic backtesting environment that better mimics real-world trading scenarios.

Our pair selection strategy combines systematic correlation-based screening with economically-motivated selection where

fundamentals provide directional guidance about cointegration properties. This dual approach enables comprehensive evaluation of our Bayesian framework across varying degrees of theoretical prior strength, from purely statistical relationships to pairs where economic theory strongly predicts $\beta_1 > 0$. This selection methodology reflects established practice in both academic research and quantitative trading, where practitioners routinely combine statistical filters with economic rationale to identify tradeable relationships (Rosenthal and Young 1990, Ungever 2015, Lee and Leung 2020, Chen and Alexiou 2025, Tadi and Witzany 2025). Additionally, economically-motivated selection inherently involves qualitative judgment about fundamental relationships. This subjectivity is not a limitation but rather a methodological feature: it generates strong theoretical priors about parameter signs that enable rigorous testing of our Bayesian framework’s capacity to identify when observed data conflicts with these expectations.

For U.S. markets, we implement both selection approaches. The systematic component computes long-term correlations for all S&P 500 constituents within each Global Industry Classification Standard (GICS) sector, retaining only pairs exceeding a minimum correlation threshold of 0.8; the two highest-correlation pairs per sector form the systematic sample, as reported in table A1 in the Appendix section. The economically-motivated component adds one pair per sector based on observable fundamental characteristics that create expectations of co-movement. These include dual-class share structures where voting and non-voting shares represent claims on identical underlying assets; firms competing directly in narrow product markets such as beverages, payment processing networks, semiconductor manufacturing, or steel production; firms sharing critical input costs and exposure to common regulatory frameworks such as utilities or real estate investment trusts; and firms with overlapping operational models in sectors such as automotive manufacturing, electrical equipment, or oilfield services. These characteristics provide economic rationale for long-term price relationships and generate strong priors for positive hedge ratios independently of observed correlations.

For the Brazilian market, structural constraints of the Ibovespa index preclude systematic sector-based selection comparable to U.S. analysis. The Ibovespa contains limited sectoral representation, with several sectors having single constituents or no representation among liquid stocks. Consequently, we focus exclusively on economically-motivated selection among Ibovespa constituents. First, we analyzed all liquid dual-class share pairs where both voting (ON) and non-voting preferred (PN) shares trade actively. Brazilian corporate law permits these structures, which represent claims on the same underlying business differing primarily in voting rights and dividend priority, with preferred shares typically receiving dividend preference in exchange for limited or no voting power. We identified seven such pairs across the financial, utilities, materials and energy sectors. Second, we included pairs of companies within the same economic group where shared ownership structures and strategic coordination suggest long-term price relationships, which yields five additional pairs. Both categories provide strong economic priors for $\beta_1 > 0$ given fundamental similarities, enabling a more

[†] In this work, we assign $q = 1/2$ and $r = 1$ as initial values for the Markov Chain - Monte Carlo simulation.

comprehensive evaluation of the prior-data conflict detection capacity of our framework.

To ensure practical implementability and avoid spurious results from thin markets, we subjected all Brazilian pairs to rigorous liquidity screening. We retained only pairs in which both shares demonstrated consistent daily trading volumes and reasonable bid-ask spreads suitable for institutional execution. This verification process excluded some dual-class pairs in which one of the shares exhibited insufficient liquidity despite theoretical cointegration properties. Retained pairs represent actively traded securities where observed pricing dislocations reflect genuine arbitrage opportunities rather than artifacts of illiquid markets with unrealistic execution assumptions.

4.2. Capital utilization

We allocated initial capital of \$10,000 per pair for U.S. markets and R\$10,000 per pair for Brazilian markets. These amounts were set uniformly across all pairs prior to any empirical analysis, avoiding look-ahead bias. Because all performance metrics are scale-invariant under proportional capital scaling, these levels do not affect relative strategy comparisons or risk-adjusted metrics.

We adopt full capital deployment with continuous reinvestment of all profits to generate compound returns. Capital is allocated between long and short positions according to formulas (6) and (7), which maintain the appropriate hedge ratio implied by the cointegration relationship while ensuring complete capital utilization. We do not impose leverage constraints or explicitly model margin requirements for short positions.

This framework applies identically to both the reference cointegration strategy and our proposed Bayesian approach, ensuring that performance differences arise solely from signal generation mechanisms rather than position sizing variations.

4.3. Sliding windows

To comprehensively test both the reference and proposed strategies, we implemented a two-stage walk-forward testing framework with nested sliding windows. The outer window partitions the data into in-sample and out-of-sample periods: we evaluate strategy hyperparameters (L, U, α, w) based on in-sample performance and select the optimal configuration for the subsequent out-of-sample trading period. The inner window (estimation window with w observations) operates within these periods, continuously updating model parameters by testing for cointegration, estimating the hedge ratio β_1 , and deriving its posterior distribution.

4.3.1. Outer sliding window. At the outer-window level, we partition the dataset into calibration and testing periods of three and one month, respectively. Under the 30-min intraday timeframe, three months provides sufficient data (approximately 60 trading days with 13 observations each) to capture meaningful performance patterns while remaining responsive to evolving market dynamics. One month offers a reasonable validation period balancing statistical power with practical

relevance. For instance, in a pair with data starting in February 2015, the first calibration window covers February–April 2015, with May 2015 serving as the test period. The process then rolls forward monthly: March–May 2015 becomes the next calibration window with June 2015 separated for testing, continuing through the entire dataset.

Within each calibration period, we evaluate combinations of entry/exit levels (L/U) , significance level (α) , and inner-window length $w \in \{65, 130\}$, selecting the configuration that maximizes the cumulative return factor. This optimal parameter set is then held fixed and applied to the subsequent test month for out-of-sample validation.

4.3.2. Inner sliding window. With strategic parameters determined at the outer level, the inner sliding window of length w progresses incrementally through the 30-min price series during both calibration and testing phases. At each step, this window tests for cointegration, estimates the hedge ratio β_1 , and derives its posterior distribution, enabling the strategy to adapt continuously to the most recent w observations while maintaining the fixed strategic configuration selected during calibration.

The inner sliding window operates in three stages: estimation, decision, and trading periods. The estimation period has length w and is used to estimate the model parameters. Assuming that the decision period in which we opt to trade or stay out of the market occurs at time t , the estimation period spans from $t - w + 1$ to $t - 1$, while the beginning of the trading period is expressed as $t + 1$. If the pair is not cointegrated in the estimation period, we slide the rolling window one discrete time unit forward and re-evaluate the cointegration status. Alternatively, if the pair is cointegrated in the estimation window, we draw random values from the full conditional of β using a Gibbs sampler and monitor the drawn sequence of β_1 values against the negative tolerance threshold m . If at any stage of the sampling process m consecutive draws of β_1 are all negative, the cointegration status is discarded for that period and the window slides forward, as described in Section 3.1. Otherwise, if the pair passes this confirmation step throughout the entire sampling process, the posterior mean of the distribution for β_1 , i.e. $\mathbb{E}[\beta_1 | \mathbf{y}, \beta_0, \sigma^2, \beta_1 > 0]$, is adopted as the hedge ratio coefficient for the subsequent trading decisions.

Using the pre-defined α (from the outer-window optimization), we obtain two quantiles q_α and $q_{1-\alpha}$ under the full conditional of β_1 , which serve as confirmatory limits for our entry and stop-loss signals. We then compute the standardized spread for the given period using the pre-defined levels $(L$ and $U)$ to trigger operations, i.e. sell the spread at U and buy the spread at L .

Once the trading parameters are defined, we proceed to the decision and trading stages. We start by fitting a linear regression model $\hat{y}_{[t-w+1:t]} = \hat{\beta}_{0,[t-w+1:t]} + \hat{\beta}_{1,[t-w+1:t]} x_{[t-w+1:t]}$ to the period comprised of the entire estimation window plus an additional data point, i.e. from $t - w + 1$ to t , and store $\hat{\beta}_{1,[t-w+1:t]}$.

If the standardized spread has crossed any of the defined boundaries (either L or U) and $\hat{\beta}_{1,[t-w+1:t]}$ is well behaved, i.e. $q_\alpha < \hat{\beta}_{1,[t-w+1:t]} < q_{1-\alpha}$, an entry signal is emitted for $t + 1$.

Once an operation is initiated in $t + 1$, we start monitoring for exit signals from $t + 2$ onward.

The take-profit signal is activated when the standardized spread reverts to its theoretical mean of 0 within subsequent periods. A stop-loss signal is triggered at time $t + 1 + k$, where $k \in \{1, 2, 3, \dots\}$, whenever the standardized spread crosses a pre-established loss margin ξ (either $L - \xi$ or $U + \xi$), and the updated hedge ratio $\hat{\beta}_{1,[t-w+1:t+1+k]}$ derived from a linear regression fitted from the start of the estimation period to the current time falls outside the specified quantiles, i.e. $\hat{\beta}_{1,[t-w+1:t+1+k]} \leq q_\alpha$ or $\hat{\beta}_{1,[t-w+1:t+1+k]} \geq q_{1-\alpha}$. This process is repeated periodically until a take-profit or stop-loss signal is emitted, and the operation is consequently terminated. Once a trade is closed, the algorithm is restarted in the next period. Algorithm 1 provides a complete summary of the inner-window walk-forward procedure.

4.4. Lot balancing

Following the standard approach in cointegration-based pairs trading literature (Vidymurthy 2004), our position sizing maintains the quantity ratio $Q_x/Q_y = \beta_1$ to replicate the stationary spread $S_t = y_t - \beta_1 x_t$ while utilizing available capital K_t . With the constraint of full capital utilization, we derive:

$$Q_{y,t+1} = \frac{K_t}{y_{t+1} + \hat{\beta}_{1,[t-w+1:t-1]} \times x_{t+1}} \quad (6)$$

$$Q_{x,t+1} = \hat{\beta}_{1,[t-w+1:t-1]} \times Q_{y,t+1} \quad (7)$$

where $Q_{y,t+1}$ and $Q_{x,t+1}$ represent the computed lot sizes for trading period $t + 1$, K_t denotes total invested capital at decision period t , $\hat{\beta}_{1,[t-w+1:t-1]}$ is the estimated hedge ratio from the estimation period, and y_{t+1} and x_{t+1} represent the execution prices of assets Y and X at entry time $t + 1$.

To simulate realistic execution conditions, we compute y_{t+1} and x_{t+1} as the average of the 5-min prices observed during a 30-min execution window following the entry signal. This approach accounts for practical trading frictions including slippage, order execution delays, and market impact, providing more realistic performance estimates than assuming instantaneous execution at signal prices. The same methodology applies to exit prices when closing positions.

These formulas ensure two key properties. First, the quantity ratio $Q_x/Q_y = \hat{\beta}_1$ guarantees that the portfolio value $Q_y \times y_t - Q_x \times x_t$ replicates the stationary spread $y_t - \beta_1 x_t$ identified by cointegration. Second, total capital deployed equals $Q_y \times y_{t+1} + Q_x \times x_{t+1} = K_t$, achieving full capital utilization.

We applied this identical methodology to both the reference cointegration strategy and our Bayesian framework across all empirical analyses. This methodological consistency ensures that any observed performance differences between strategies arise exclusively from signal generation mechanisms rather than from position sizing variations.

4.5. Parameter specification and automated selection

Our model requires the specification of several parameters, which can be categorized into two groups. The first group includes parameters common to both our Bayesian approach

and the standard cointegration strategy. These parameters include: the estimation window length (w), which determines the historical period used for estimating the hedge ratio; the standardized spread thresholds (L and U), which define entry and exit conditions for trades in the first layer; and the stop-loss margin (ξ), which prevents excessive losses by triggering an exit when the spread exceeds a predefined limit. For this study, we fix $\xi = 1$ across all experiments to maintain consistency and focus our analysis on the other strategic parameters. This choice provides balanced risk control: with entry thresholds of $L/U = \pm 1$ or ± 2 standard deviations, setting $\xi = 1$ creates stop-loss boundaries at ± 2 or ± 3 standard deviations respectively, corresponding to risk-reward ratios of 1:1 and 2:1 that align with standard risk management practices in pairs trading. The take-profit margin is implicitly defined at the equilibrium point (0), ensuring that positions are closed when the spread reverts to its historical mean. As described in Section 4.3.1, the parameters w , L , U , and α are optimized at the outer-window level using the three-month calibration period to maximize the cumulative return factor, and the optimal combination is then applied to the subsequent one-month test period.

The second group consists of a single parameter unique to the Bayesian framework: the quantile parameter (α), which has a key role in defining probabilistic entry and exit signals. Unlike the standard approach, which relies on fixed spread levels, our Bayesian method utilizes the posterior distribution of the hedge ratio to dynamically adjust trade execution. The quantile parameter determines the degree of certainty required before executing a trade, thereby influencing both trading frequency and risk exposure. This parameter is also optimized during the outer-window calibration process.

Rather than imposing a single parameter configuration across all pairs, we provide two distinct candidate values for each strategic parameter: $w \in \{65, 130\}$, $L/U \in \{\pm 1, \pm 2\}$, and $\alpha \in \{0.05, 0.15\}$. The in-sample/out-of-sample framework then automatically selects the optimal combination from the grid for each individual pair based on cumulative return factor performance during the three-month calibration period. This pair-specific optimization occurs dynamically at every outer-window step: as the sliding windows advance month by month, the model continuously re-evaluates all parameter combinations and updates the optimal configuration for each pair independently. Consequently, the same pair may operate with different parameter settings across different calibration periods as market conditions evolve, ensuring that the strategy adapts to changing pair dynamics without manual intervention. This automated, adaptive approach closely mirrors real-world trading practice, where practitioners back-test multiple configurations and deploy the best-performing setup, periodically recalibrating as new data becomes available. This automated and adaptive approach is robust and free from look-ahead bias, as each test period is evaluated exclusively under parameters optimized on prior data.

The following subsections describe the parameter space provided to the automated selection framework and discuss the observed patterns in optimal configurations selected across pairs and time periods. These observations reflect the aggregated results of the pair-specific dynamic optimization process rather than predetermined parameter choices.

Algorithm 1 Inner-window parameter estimation and trading signal generation

Require: Time series observations $\{(x_t, y_t)\}_{t=1}^n$, sliding window length w , MCMC replications M , negative $\hat{\beta}_1$ tolerance m , lower bound entry threshold L , upper bound entry threshold U , stop-loss margin ξ , quantile parameter α

- 1: $\text{flag} \leftarrow \text{FALSE}$ ▷ Flag indicating the existence of an active operation
- 2: $\text{side} \leftarrow \text{NULL}$ ▷ Side of entry: *upper* if $z \geq U$, *lower* if $z \leq L$
- 3: **for** t in $[(w + 1) : n]$ **do**
- 4: **if** flag is **FALSE** **then**
- 5: Run cointegration test on data $\{(x_s, y_s)\}_{s=t-w+1}^{t-1}$
- 6: **if** pair is cointegrated **then**
- 7: Run Gibbs sampler on data $\{(x_s, y_s)\}_{s=t-w+1}^{t-1}$ and obtain the full conditional $\beta | y, \sigma^2$
- 8: **if** $\beta_1^{(i)} < 0$ for some m consecutive $i \in \{1, \dots, M\}$ **then** ▷ m consecutive negative β_1 draws detected
- 9: Discard the cointegration status for the period ▷ Pair selection filter
- 10: **else**
- 11: Compute $\hat{\beta}_{1,[t-w+1:t-1]} = \mathbb{E}[\beta_1 | y, \beta_0, \sigma^2, \beta_1 > 0]$
- 12: Compute q_α and $q_{1-\alpha}$
- 13: Compute spread $\delta_{[t-w+1:t-1]} = y_{[t-w+1:t-1]} - \hat{\beta}_{1,[t-w+1:t-1]} x_{[t-w+1:t-1]}$ on data $\{(x_s, y_s)\}_{s=t-w+1}^{t-1}$
- 14: Compute standardized spread at time t :
- 15: $z[t] = \frac{\delta[t] - \delta_{[t-w+1:t-1]}}{s_{\delta_{[t-w+1:t-1]}}}$
- 16: Fit $\hat{y}_{[t-w+1:t]} = \hat{\beta}_{0,[t-w+1:t]} + \hat{\beta}_{1,[t-w+1:t]} x_{[t-w+1:t]}$ to data $\{(x_s, y_s)\}_{s=t-w+1}^t$
- 17: Store $\hat{\beta}_{1,[t-w+1:t]}$
- 18: **if** $z[t] \geq U$ and $q_\alpha < \hat{\beta}_{1,[t-w+1:t]} < q_{1-\alpha}$ **then** ▷ Upper entry trigger
- 19: ENTRY Signal on $t + 1$
- 20: $\text{flag} \leftarrow \text{TRUE}$, $\text{side} \leftarrow \text{upper}$
- 21: **else if** $z[t] \leq L$ and $q_\alpha < \hat{\beta}_{1,[t-w+1:t]} < q_{1-\alpha}$ **then** ▷ Lower entry trigger
- 22: ENTRY Signal on $t + 1$
- 23: $\text{flag} \leftarrow \text{TRUE}$, $\text{side} \leftarrow \text{lower}$
- 24: **end if**
- 25: **end if**
- 26:
- 27: **end if**
- 28: **else** ▷ Look for exit triggers
- 29:
- 30: Fit $\hat{y}_{[t-w+1:t+1+k]} = \hat{\beta}_{0,[t-w+1:t+1+k]} + \hat{\beta}_{1,[t-w+1:t+1+k]} x_{[t-w+1:t+1+k]}$ to data $\{(x_s, y_s)\}_{s=t-w+1}^{t+1+k}$ where $t + 1 + k$ represents current time
- 31: Store new $\hat{\beta}_{1,[t-w+1:t+1+k]}$
- 32: Compute spread $\delta_{[t-w+1:t+1+k]} = y_{[t-w+1:t+1+k]} - \hat{\beta}_{1,[t-w+1:t+1+k]} x_{[t-w+1:t+1+k]}$ on data $\{(x_s, y_s)\}_{s=t-w+1}^{t+1+k}$
- 33: Compute standardized spread at current time $t + 1 + k$:
- 34: $z[t + 1 + k] = \frac{\delta[t + 1 + k] - \delta_{[t-w+1:t+1+k]}}{s_{\delta_{[t-w+1:t+1+k]}}}$
- 35: **if** $(z[t + 1 + k] \leq L - \xi \text{ or } z[t + 1 + k] \geq U + \xi)$ and $(\hat{\beta}_{1,[t-w+1:t+1+k]} \leq q_\alpha \text{ or } \hat{\beta}_{1,[t-w+1:t+1+k]} \geq q_{1-\alpha})$ **then**
- 36: EXIT Signal ▷ Stop-loss signal
- 37: $\text{flag} \leftarrow \text{FALSE}$
- 38: **else if** $(\text{side} = \text{upper} \text{ and } z[t + 1 + k] \leq 0) \text{ or } (\text{side} = \text{lower} \text{ and } z[t + 1 + k] \geq 0)$ **then**
- 39: EXIT Signal ▷ Take-profit signal
- 40: $\text{flag} \leftarrow \text{FALSE}$
- 41: **end if**
- 42: **end if**
- 43: **end for**

4.5.1. Estimation window (w). For intraday data in the 30-min timeframe, we provided estimation window lengths of 65 and 130 observations, representing approximately two and four trading weeks of data, respectively. These values were chosen because they provide sufficient data points for robust statistical analysis while remaining responsive to recent market conditions in the underlying timeframe.

The automated parameter selection across calibration periods revealed no consistent advantage between using $w = 65$ or $w = 130$ across all pairs. Optimal window lengths varied substantially between pairs and time periods, with the outer-window optimization frequently alternating between both values for the same pair as market conditions evolved. This variation likely reflects differences in individual pairs'

underlying price dynamics and cointegration characteristics, which is why the pair-specific optimization process selects the appropriate window length for each specific context.

The absence of a universal optimal window length aligns with theoretical expectations. Excessively small windows can increase the chances of an incorrect rejection of the cointegration hypothesis (Type II errors) due to low statistical power, higher estimator variance, and insufficient capture of long-run relationships (Cheung and Lai 1993, Otero and Smith 2000). In theory, larger samples should improve cointegration detection by increasing statistical power; however, in practice, excessively large time series tend to result in a rejection of cointegration status due to the inclusion of structural breaks or regime changes that disrupt long-term equilibrium relationships (Hakkio and Rush 1991, Gregory and Hansen 1996). This dual challenge in window selection has been well documented in the econometric literature (Engle and Granger 1987, Johansen 1988, 1991), highlighting the importance of finding an appropriate balance in window sizing: large enough to ensure statistical robustness but constrained enough to avoid capturing multiple structural changes that could mask existing cointegrating relationships.

4.5.2. Standardized spread thresholds (L and U). We provided thresholds of ± 1 and ± 2 standard deviations of the standardized spread. The ± 1 standard deviation thresholds represent a moderately aggressive approach, capturing deviations that occur with reasonable frequency while maintaining some selectivity. The ± 2 standard deviation thresholds represent a more conservative approach, targeting only larger deviations that are more likely to represent genuine mispricing, though at the cost of reduced signal frequency. These two values provide a balanced range for assessing how threshold selection affects strategy performance across different market conditions and pair characteristics.

The automated selection process predominantly favored thresholds of ± 1 standard deviation for both the reference and the proposed strategy across most pairs and periods. The prevalence of ± 1 as the optimal threshold likely reflects the nature of high-frequency intraday data, where mean-reverting opportunities tend to be smaller in magnitude but occur more frequently, making tighter thresholds more suitable for capturing profitable trades. While ± 2 thresholds were occasionally selected as optimal for certain pairs during specific windows, the overall pattern suggests that moderately aggressive entry levels are generally more appropriate for 30-minute intraday pair trading. This finding aligns with the practical observation that intraday strategies require responsiveness to capture short-lived arbitrage opportunities before they dissipate.

4.5.3. Quantile parameter (α). The quantile parameter α exerts significant control over the responsiveness of the proposed model. This parameter effectively regulates the occurrence of entry and stop-loss signals in the following manner: a smaller α value (e.g. 0.05) expands the area in which β_1 is assumed to be well-behaved, making the Bayesian method less selective for entry signals and more selective for stop-loss signals. Conversely, a larger α value (e.g. 0.15) creates a narrower middle band, which filters more potential entries

compared to the reference method, but makes the algorithm less selective and thus more responsive to exit signals from the standard cointegration layer (figure 1).

We provided two quantile levels: $\alpha \in \{0.05, 0.15\}$. The automated selection process revealed no dominant α value across all pairs. Optimal performance was well distributed between the two tested levels, with the outer-window optimization frequently switching between $\alpha = 0.05$ and $\alpha = 0.15$ for the same pair across different time periods depending on evolving market conditions. This heterogeneity suggests that the appropriate level of selectivity is pair-dependent and reflects differences in underlying volatility patterns, cointegration strength, and noise characteristics. This variation underscores the importance of the automated optimization process in identifying the most suitable quantile parameter for each pair-specific context.

5. Results

The empirical results presented in tables 1 and 2 demonstrate that the Bayesian strategy outperforms the traditional cointegration approach in the majority of analyzed pairs across both U.S. and Brazilian markets, though with markedly different magnitudes and consistency patterns. Our proposed method systematically exhibits enhanced risk-adjusted performance metrics, characterized by higher Sharpe and Sortino ratios in most configurations. Remarkably, these improvements are achieved while executing, on average, 24% fewer trades than the reference strategy, a reduction that simultaneously enhances profitability and decreases transaction costs. These findings provide strong empirical support for the hypothesis outlined in Section 3.1 regarding the enhanced precision of distribution-based trading signals.

The performance patterns differ considerably between markets, reflecting their distinct efficiency characteristics. U.S. equity markets demonstrate consistent but modest improvements, with the Bayesian method prevailing in approximately 62% of analyzed pairs despite constrained absolute return potential. In contrast, Brazilian markets exhibit both higher absolute returns and more pronounced advantages for the Bayesian approach in dual-class share pairs (voting vs. non-voting) where structural characteristics create favorable conditions for statistical arbitrage. These contrasting outcomes provide insights into the methodology's adaptability across market efficiency regimes and its relative value proposition in developed versus emerging market contexts.

5.1. U.S. market performance patterns

Table 1 organizes U.S. equity pairs by Global Industry Classification Standard (GICS) sector and selection methodology (systematic vs. economic), enabling structured evaluation of the Bayesian framework's effectiveness across different market segments. The systematic selection process identified pairs with highest correlations within each sector, while economic interest pairs were selected based on operational similarity where prior beliefs about cointegration relationships ($\beta_1 > 0$) have stronger theoretical foundations.

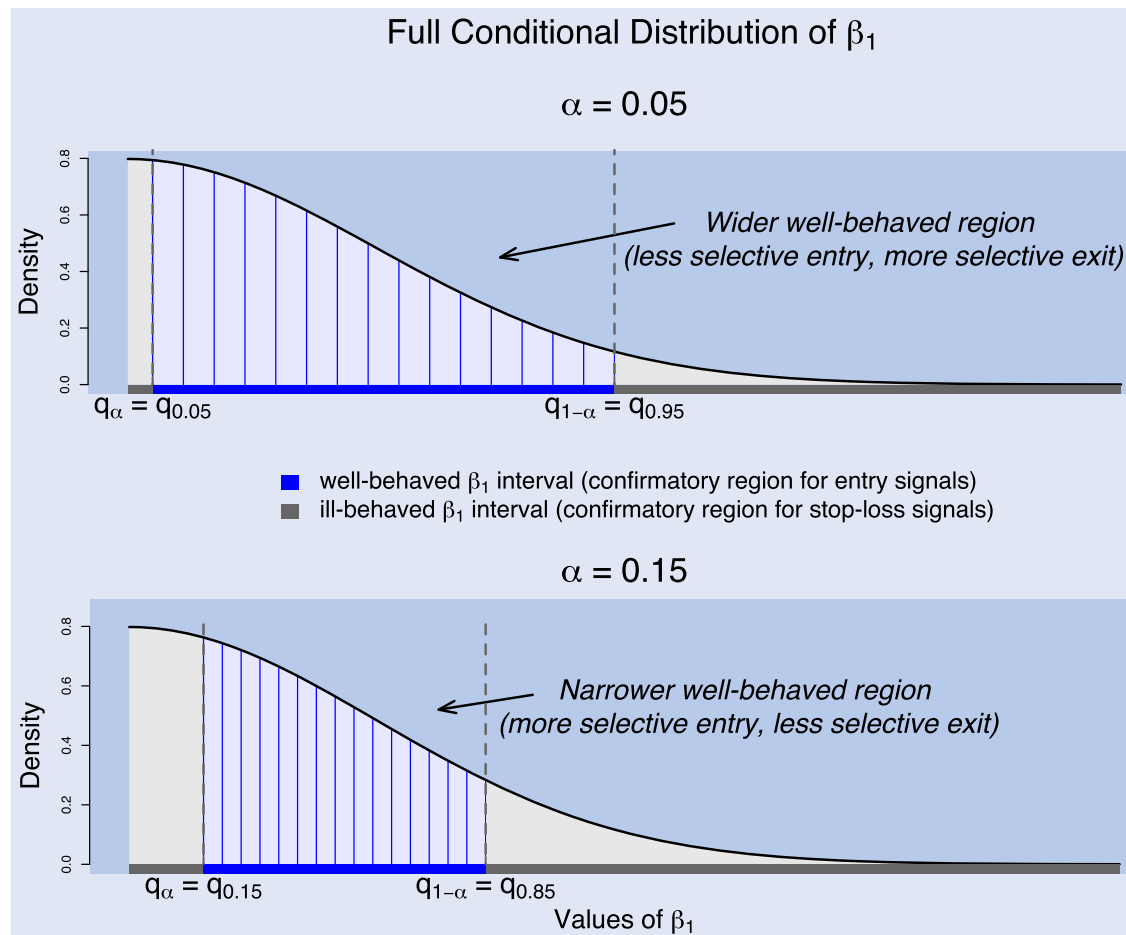


Figure 1. Quantile parameter α and ‘well-behaved’ intervals for β_1 under the full conditional distribution. The upper and lower panels illustrate how different values of α affect the confirmatory regions for entry and exit signals. For $\alpha = 0.05$ (upper panel), the well-behaved region is wider, resulting in less selectivity for entry signals and more selectivity for exit signals. For $\alpha = 0.15$ (lower panel), the well-behaved region is narrower, resulting in more selectivity for entry signals and less selectivity for exit signals. The blue regions represent intervals where β_1 is considered well-behaved, while the gray regions indicate intervals where β_1 falls outside the quantiles defined in the full conditional distribution.

The Bayesian approach outperformed the reference strategy in 18 of 29 analyzed U.S. equity pairs (62%). While absolute returns remained modest for many pairs, with several exhibiting cumulative return factors below 1.15 over the ten-year period, the pattern of relative improvement demonstrates the value of Bayesian signal confirmation. These outcomes reflect efficiency constraints characteristic of developed equity markets that fundamentally limit statistical arbitrage opportunities. The primary value of the Bayesian framework in this context lies in its relative improvement through enhanced signal precision and risk management, rather than in generating commercially exceptional absolute returns.

5.1.1. Performance contrast analysis. Figure 2 presents a contrast analysis organized by absolute performance differences between methods. The top two rows show the six pairs that exhibit the largest absolute differences in cumulative return factors, while the bottom two rows present pairs where both methods converge to nearly identical outcomes. This organization reveals where methodological choice matters the most versus where market efficiency constrains both approaches similarly.

The five pairs exhibiting largest absolute differences favoring the Bayesian method, in percentage points, are: AMAT-KLAC (27.9pp), GM-F (16.4pp), LEN-DHI (14.1pp), DOW-LYB (11.6pp), and PEP-KO (9.0pp). The sixth pair with the largest absolute difference, HAL-SLB (16.3pp), represents a notable exception where the reference strategy substantially outperformed.

The AMAT-KLAC pair (Information Technology, systematic selection) exhibited the largest differential favoring Bayesian approach, with cumulative return factor of 1.32950 versus 1.05019, accompanied by substantial Sharpe improvement from 0.17059 to 0.73805 and Sortino enhancement from 0.25203 to 1.16372, while simultaneously reducing maximum drawdown from 0.08539 to 0.05555. These improvements were achieved with 162 trades versus 190 for the reference method, representing a 15% reduction in trading activity. The GM-F pair (Consumer Discretionary, economic) achieved 1.29090 versus 1.12730, with Sharpe improvement from 0.33599 to 0.49544 and Sortino enhancement from 0.50032 to 1.03483, executing only 102 trades compared to the reference’s 140 (27% reduction).

The LEN-DHI pair (Consumer Discretionary, systematic) demonstrated substantial improvement from 0.96799

Table 1. Comparison between the Bayesian approach ('Bayes') and the traditional cointegration strategy ('Ref.') organized by Global Industry Classification Standard (GICS) sector and selection type (Systematic vs. Economic).

		CRF		Trades		Volatility		Max. DD		Sharpe		Sortino	
Selection	Pair	Ref.	Bayes	Ref.	Bayes	Ref.	Bayes	Ref.	Bayes	Ref.	Bayes	Ref.	Bayes
Equities													
Communication services													
Systematic	GOOGL-GOOG	1.03400	1.06068	876	717	0.00673	0.00569	0.01848	0.01028	0.51942	1.07960	0.70221	1.78759
Systematic	FOXA-FOX	1.02584	1.06296	726	528	0.00939	0.01075	0.02405	0.01715	0.29469	0.60427	0.37461	0.80264
Economic	NWSA-NWS	1.14614	1.17280	974	783	0.01061	0.01183	0.01234	0.01643	1.33585	1.39911	2.34496	2.22259
Consumer discretionary													
Systematic	LEN-DHI	0.96799	1.10873	207	159	0.01866	0.02534	0.07461	0.03972	-0.17823	0.45025	-0.24648	0.95557
Systematic	MAR-HLT	0.90461	0.92438	207	174	0.02662	0.02156	0.12782	0.09270	-0.38901	-0.37548	-0.46706	-0.48889
Economic	GM-F	1.12730	1.29090	140	102	0.04106	0.06374	0.07694	0.09682	0.33599	0.49544	0.50032	1.03483
Consumer staples													
Economic	PEP-KO	1.02071	1.11089	160	125	0.01520	0.02161	0.04711	0.02297	0.15386	0.53886	0.23484	0.96138
Energy													
Systematic	COP-EOG	1.01425	0.98498	137	111	0.01698	0.01945	0.03720	0.05692	0.09526	-0.07138	0.15798	-0.10339
Systematic	MPC-VLO	0.97766	0.98798	202	143	0.03456	0.03562	0.10735	0.13112	-0.05074	-0.01739	-0.05810	-0.01997
Economic	HAL-SLB	1.24799	1.08483	175	140	0.03730	0.04217	0.03264	0.09997	0.63871	0.22234	1.97366	0.39573
Financials													
Systematic	CFG-FITB	1.00188	0.94301	167	128	0.01992	0.02044	0.05413	0.11286	0.01985	-0.29249	0.02900	-0.39047
Systematic	MET-PRU	0.96885	0.96980	207	149	0.01598	0.02135	0.06756	0.07236	-0.19712	-0.13810	-0.29513	-0.18116
Economic	V-MA	1.04185	1.00685	227	162	0.01981	0.01765	0.04209	0.02964	0.22815	0.04952	0.41767	0.07913
Healthcare													
Economic	A-WAT	0.93617	0.93285	186	146	0.03078	0.02548	0.11921	0.11458	-0.21973	-0.29039	-0.26703	-0.36996
Industrials													
Systematic	RSG-WM	0.99823	0.99979	292	222	0.01251	0.01411	0.03918	0.04041	-0.00847	0.00547	-0.01200	0.00710
Systematic	ADP-PAYX	1.03103	1.04758	174	131	0.01820	0.02309	0.03761	0.04324	0.18620	0.22376	0.30358	0.36160
Economic	ETN-PWR	1.14482	1.15912	131	99	0.02822	0.02765	0.03557	0.04385	0.52024	0.58134	1.21392	1.07830
Information technology													
Systematic	KLAC-LRCX	1.26881	1.20968	207	167	0.05808	0.06211	0.09535	0.10993	0.46046	0.35282	0.77020	0.51532
Systematic	AMAT-KLAC	1.05019	1.32950	190	162	0.03330	0.04185	0.08539	0.05555	0.17059	0.73805	0.25203	1.16372
Economic	AMD-NVDA	0.76135	0.67188	94	80	0.05270	0.05006	0.25226	0.33069	-0.51923	-0.81098	-0.72146	-0.91306
Materials													
Systematic	DOW-LYB	0.90906	1.02521	94	71	0.02465	0.01694	0.11762	0.03310	-0.71586	0.28962	-0.78292	0.43521
Systematic	MLM-VMC	1.00221	0.96615	309	193	0.03028	0.03160	0.07370	0.09105	0.02566	-0.14361	0.03850	-0.19650
Economic	NUE-STLD	1.06113	0.98060	194	154	0.02849	0.03260	0.08426	0.11324	0.23356	-0.04604	0.33924	-0.06260
Real estate													
Systematic	FRT-REG	0.97615	1.02436	264	236	0.02596	0.02956	0.10136	0.05466	-0.08390	0.09973	-0.10570	0.12401
Systematic	EQR-UDR	0.97135	0.90445	234	180	0.01623	0.01850	0.04702	0.10225	-0.17918	-0.55138	-0.26592	-0.69657
Economic	AVB-EQR	1.09348	1.08187	288	241	0.02643	0.03234	0.06326	0.05441	0.36764	0.27158	0.57858	0.37966
Utilities													
Systematic	CMS-DTE	0.97001	1.03045	215	151	0.01521	0.01478	0.05979	0.02589	-0.20671	0.22114	-0.26459	0.38771
Systematic	CMS-WEC	0.98951	1.04953	229	168	0.00972	0.01238	0.03687	0.02165	-0.10782	0.41655	-0.16715	0.63811
Economic	SO-NEE	0.94307	0.95100	138	110	0.01431	0.01581	0.06765	0.07755	-0.42792	-0.32979	-0.65914	-0.44336

Note: Results based on cumulative return factor (CRF), trade count, volatility, maximum drawdown (Max. DD), Sharpe, and Sortino ratios from 02/01/2015 to 01/31/2025 in the U.S. market.

to 1.10873, transforming a losing strategy into a profitable one while dramatically improving Sharpe ratio from -0.17823 to 0.45025 and Sortino from -0.24648 to 0.95557 , accompanied by substantial maximum drawdown reduction from 0.07461 to 0.03972 . The Bayesian method achieved these results with 159 trades versus 207 for the reference (23% reduction). The DOW-LYB pair (Materials, systematic) showed considerable improvement from 0.90906 to 1.02521 , with Sharpe transformation from -0.71586 to 0.28962 and dramatic maximum drawdown reduction from 0.11762 to 0.03310 , executing only 71 trades versus 94 (24% reduction). The PEP-KO pair (Consumer Staples, economic) reached 1.11089 versus 1.02071 with enhanced Sharpe from 0.15386 to 0.53886 and improved Sortino from 0.23484 to 0.96138 , while reducing trades from 160 to 125 (22% reduction).

The HAL-SLB pair (Energy, economic) represents the most significant case where the reference method outperformed, achieving 1.24799 versus 1.08483 with superior Sharpe (0.63871 vs. 0.22234) and dramatically better Sortino (1.97366 vs. 0.39573), though the Bayesian method still executed 20% fewer trades (140 vs. 175).

Conversely, the six pairs with smallest absolute differences demonstrate configurations where methodological choice has minimal impact. These include: MET-PRU (0.1pp, Bayesian favored), RSG-WM (0.2pp, Bayesian favored), A-WAT (0.3pp, reference favored), SO-NEE (0.8pp, Bayesian favored), MPC-VLO (1.0pp, Bayesian favored), and AVB-EQR (1.2pp, reference favored). Even in these convergent cases, the Bayesian method consistently executed fewer trades: MET-PRU (149 vs. 207, 28% reduction), RSG-WM (222 vs. 292, 24% reduction), A-WAT (146 vs. 186, 22%

Table 2. Comparison between the Bayesian approach ('Bayes') and the traditional cointegration strategy ('Ref.').

		CRF		Trades		Volatility		Max. DD		Sharpe		Sortino	
Selection	Pair	Ref.	Bayes	Ref.	Bayes	Ref.	Bayes	Ref.	Bayes	Ref.	Bayes	Ref.	Bayes
Voting vs. Non-Voting shares													
Economic	ITUB3-ITUB4	1.28718	1.39006	951	689	0.01886	0.02055	0.01226	0.01401	3.12032	3.73766	5.73131	6.20949
Economic	BBDC3-BBDC4	1.19170	1.18425	534	419	0.01787	0.02109	0.01470	0.01935	2.28352	1.87405	4.40451	2.96109
Economic	ELET3-ELET6	1.20296	1.47916	613	461	0.02550	0.02827	0.02765	0.01371	1.73360	3.24929	3.29522	6.38872
Economic	GGBR3-GGBR4	1.67260	1.84105	1255	868	0.04425	0.04483	0.02782	0.02319	2.71535	3.17883	4.60221	5.26139
Economic	PETR3-PETR4	1.13932	1.16663	482	377	0.02048	0.02217	0.02952	0.02079	1.49323	1.62987	2.64396	2.74771
Economic	USIM3-USIM5	1.48332	1.56613	776	619	0.03880	0.04239	0.06071	0.03241	2.36004	2.45928	3.98479	3.90737
Economic	POMO3-POMO4	2.26305	2.97407	997	791	0.06187	0.06201	0.05097	0.03686	3.40397	4.52671	5.93563	8.13925
Same economic group													
Economic	ITUB4-ITSA4	1.20107	1.18405	371	302	0.03879	0.04972	0.04603	0.05459	1.11838	0.81541	2.06948	1.28473
Economic	UGPA3-VBBR3	1.13011	1.05270	158	115	0.04390	0.04289	0.04586	0.06111	0.75628	0.33686	1.39431	0.50240
Economic	GGBR4-GOAU4	1.10224	1.08640	682	501	0.05220	0.06264	0.07346	0.10123	0.45666	0.33698	0.70316	0.46213
Economic	CBAV3-NEOE3	1.04810	0.98314	75	40	0.08972	0.03750	0.11818	0.05669	0.20757	-0.13342	0.31007	-0.17931
Economic	CSAN3-RAIZ4	1.14176	1.16735	113	76	0.06397	0.07996	0.05963	0.07301	0.70159	0.66534	1.36255	1.14130

Note: All pairs are selected based on economic interest (dual-class shares or same economic group). Results based on cumulative return factor (CRF), trade count, volatility, maximum drawdown, Sharpe, and Sortino ratios from 06/22/2020 to 01/31/2025 in the Brazilian market.

reduction), SO-NEE (110 vs. 138, 20% reduction), MPC-VLO (143 vs. 202, 29% reduction), and AVB-EQR (241 vs. 288, 16% reduction). The A-WAT pair (Healthcare, economic) represents a case where the reference method achieved marginal superiority (0.93617 vs. 0.93285), though performance differences remain minimal with both methods generating modest losses. The AVB-EQR pair (Real Estate, economic) similarly shows slight reference advantage (1.09348 vs. 1.08187), with both methods near profitability thresholds.

This contrast pattern reveals that absolute differences manifest most strongly in pairs exhibiting substantial volatility where posterior distribution-based signal filtering enhances risk-adjusted performance (AMAT-KLAC, LEN-DHI, DOW-LYB), or in moderately profitable pairs where the Bayesian method amplifies returns through improved signal timing (GM-F, PEP-KO). Conversely, convergence occurs in stable pairs (MET-PRU, RSG-WM) where both methods generate similar near-break-even outcomes. The presence of cases where the reference method performs better (HAL-SLB, A-WAT, AVB-EQR) confirms absence of systematic bias in the comparative evaluation.

5.1.2. Sector-Specific performance characteristics. Across GICS sectors, the Bayesian approach demonstrated heterogeneous performance relative to the reference strategy.[†] In Communication Services, all three pairs exhibited consistent Bayesian superiority: dual-class shares GOOGL-GOOG (1.06068 vs. 1.03400) achieved Sharpe improvement from 0.51942 to 1.07960 while reducing trades from 876 to 717 (18% reduction), FOXA-FOX (1.06296 vs. 1.02584) demonstrated enhanced Sharpe from 0.29469 to 0.60427 with substantial trade reduction from 726 to 528 (27% reduction), and NWSA-NWS (1.17280 vs. 1.14614) showed Sharpe enhancement from 1.33585 to 1.39911 alongside improved Sortino ratios while executing 783 versus 974 trades (20% reduction).

Consumer Discretionary pairs consistently favored the Bayesian approach across all three analyzed pairs. LEN-DHI and GM-F demonstrated substantial improvements as discussed previously, while MAR-HLT showed modest CRF improvement from 0.90461 to 0.92438, though both methods generated losses with slightly improved Sharpe ratio (from -0.38901 to -0.37548), executing 174 versus 207 trades (16% reduction). The Consumer Staples sector, represented solely by PEP-KO, achieved strong Bayesian performance as detailed above.

In the Energy sector, results were mixed. The MPC-VLO pair showed minimal difference (0.98798 vs. 0.97766), with both methods near break-even, though the Bayesian approach executed 29% fewer trades (143 vs. 202). The COP-EOG pair slightly favored reference (1.01425 vs. 0.98498) while still reducing trades from 137 to 111 (19% reduction), while HAL-SLB strongly favored reference as discussed previously.

Information Technology pairs presented dramatic contrasts. AMAT-KLAC achieved the largest improvement across all sectors as detailed previously. However, KLAC-LRCX showed reference outperformance (1.26881 vs. 1.20968) with Sharpe decline from 0.46046 to 0.35282, though executing fewer trades (167 vs. 207, 19% reduction), while AMD-NVDA demonstrated substantial underperformance for both methods (reference: 0.76135, Bayesian: 0.67188), with both generating losses and the Bayesian method exhibiting even lower Sharpe (-0.81098 vs. -0.51923), though with 15% fewer trades (80 vs. 94).

Financials demonstrated high variance in outcomes. The CFG-FITB pair showed Bayesian underperformance (0.94301 vs. 1.00188), with both methods near break-even, though reducing trades from 167 to 128 (23% reduction). MET-PRU demonstrated minimal difference as discussed above, while V-MA represented a marginal case where reference slightly outperformed (1.04185 vs. 1.00685), with the Bayesian method executing 29% fewer trades (162 vs. 227). The Healthcare sector, represented solely by A-WAT, showed minimal difference with slight reference advantage as noted previously.

[†] See figure A1 in the Appendix for the historical cumulative return factors of the remaining U.S. pairs.

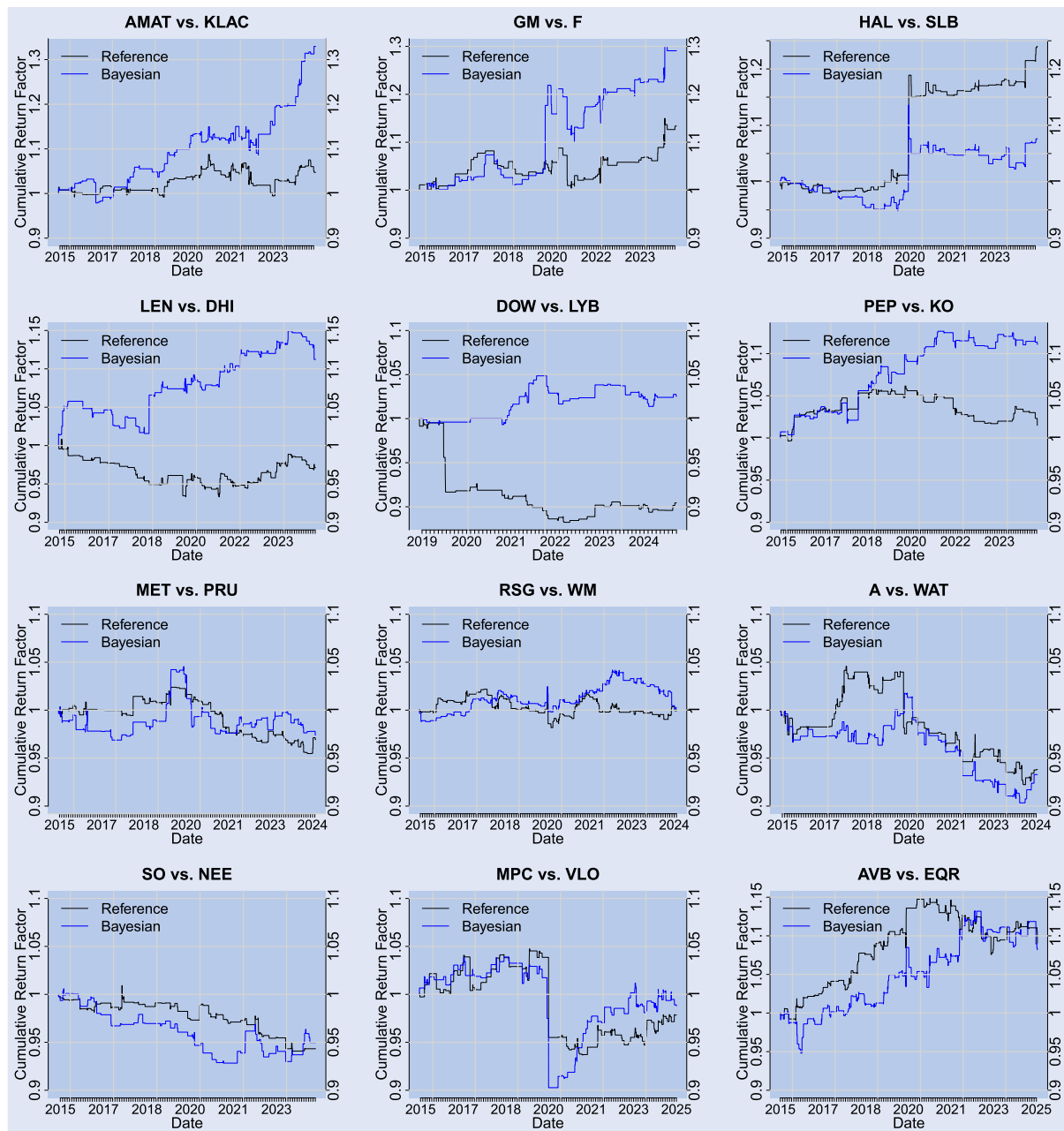


Figure 2. Cumulative return factor contrast analysis for US equity pairs from 02/01/2015 to 01/31/2025. Top two rows: pairs with largest absolute differences in percentage points between methods (AMAT-KLAC: 27.9pp Bayes favored, GM-F: 16.4pp Bayes favored, HAL-SLB: 16.3pp Ref favored, LEN-DHI: 14.1pp Bayes favored, DOW-LYB: 11.6pp Bayes favored, PEP-KO: 9.0pp Bayes favored). Bottom two rows: pairs with smallest absolute differences where methods converge (MET-PRU: 0.1pp Bayes favored, RSG-WM: 0.2pp Bayes favored, A-WAT: 0.3pp Ref favored, SO-NEE: 0.8pp Bayes favored, MPC-VLO: 1.0pp Bayes favored, AVB-EQR: 1.2pp Ref favored). Blue curves represent the Bayesian method, black curves show the reference strategy.

Materials sector pairs exhibited mixed results. DOW-LYB demonstrated substantial Bayesian advantage as detailed above. However, both MLM-VMC (0.96615 vs. 1.00221) and NUE-STLD (0.98060 vs. 1.06113) favored the reference strategy, with NUE-STLD showing Sharpe decline from 0.23356 to -0.04604 , though both pairs showed significant trade reductions: MLM-VMC from 309 to 193 (38% reduction) and NUE-STLD from 194 to 154 (21% reduction).

In Industrials, all three pairs favored the Bayesian approach. RSG-WM showed minimal difference as discussed, ADP-PAYX demonstrated modest improvement (1.04758 vs. 1.03103) with slight Sharpe enhancement (0.22376 vs. 0.18620) while reducing trades from 174 to 131 (25%

reduction), while ETN-PWR achieved improvement from 1.14482 to 1.15912 with enhanced Sharpe from 0.52024 to 0.58134 and substantial trade reduction from 131 to 99 (24% reduction).

Real Estate demonstrated mixed outcomes. FRT-REG achieved improvement from 0.97615 to 1.02436 with Sharpe transformation from -0.08390 to 0.09973 while executing 236 versus 264 trades (11% reduction). However, EQR-UDR strongly favored reference (0.97135 vs. 0.90445) with substantial Sharpe decline from -0.17918 to -0.55138 , though reducing trades from 234 to 180 (23% reduction), while AVB-EQR showed minimal difference with slight reference advantage as noted.

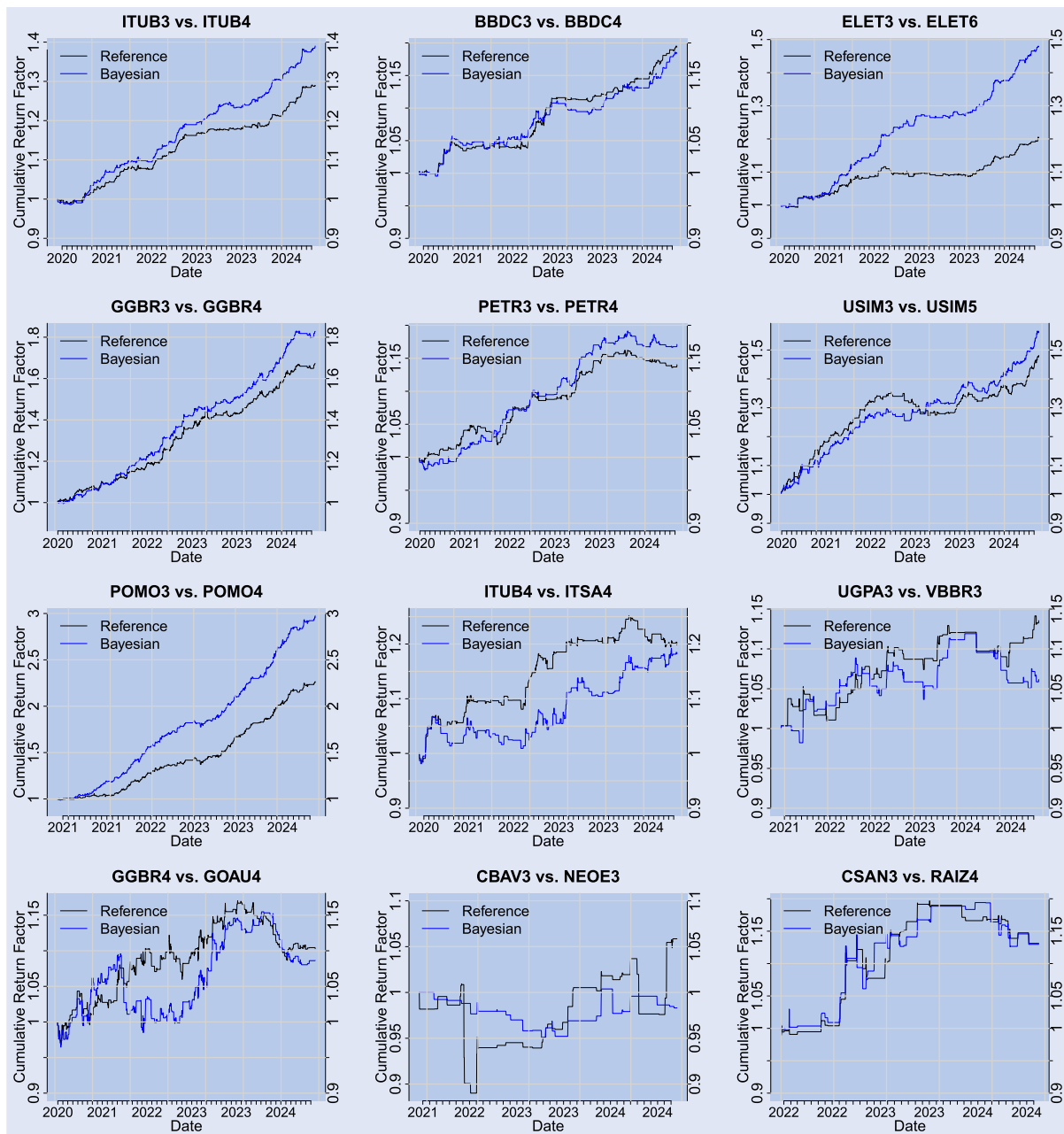


Figure 3. Cumulative return factors for all analyzed Brazilian pairs from 06/22/2020 to 01/31/2025. Blue curves represent the Bayesian method, black curves show the reference strategy.

In Utilities, two of three pairs favored the Bayesian approach. CMS-DTE achieved improvement from 0.97001 to 1.03045 with dramatically improved Sharpe (0.22114 vs. -0.20671) while reducing trades from 215 to 151 (30% reduction), while CMS-WEC showed gains from 0.98951 to 1.04953 with Sharpe enhancement from -0.10782 to 0.41655 and trade reduction from 229 to 168 (27% reduction). However, SO-NEE showed minimal difference with marginal Bayesian advantage as discussed.

5.2. Brazilian market performance

5.2.1. Voting vs. Non-Voting share pairs. Brazilian dual-class share pairs demonstrated predominantly superior performance under the Bayesian framework across six of

seven analyzed pairs, as shown in figure 3. The POMO3-POMO4 pair (Marcopolo) achieved the highest absolute returns among Brazilian pairs, with cumulative return factor of 2.97407 versus 2.26305 for the reference strategy, accompanied by superior Sharpe (4.52671 vs. 3.40397) and Sortino (8.13925 vs. 5.93563) ratios alongside reduced maximum drawdown (0.03686 vs. 0.05097), while executing 791 versus 997 trades (21% reduction).

The GGBR3-GGBR4 pair (Gerdau) demonstrated strong performance with cumulative return factor of 1.84105 versus 1.67260, achieving Sharpe ratio of 3.17883 compared to 2.71535 with reduced maximum drawdown (0.02319 vs. 0.02782) and substantial trade reduction from 1255 to 868 (31% reduction). The ELET3-ELET6 pair (Eletrobras) showed dramatic outperformance (1.47916 vs. 1.20296) with

Sharpe improvement from 1.73360 to 3.24929 and Sortino enhancement from 3.29522 to 6.38872, accompanied by substantial maximum drawdown reduction from 0.02765 to 0.01371, while reducing trades from 613 to 461 (25% reduction). The USIM3-USIM5 pair (Usiminas) achieved 1.56613 versus 1.48332 with Sharpe improvement from 2.36004 to 2.45928, with reduced maximum drawdown (0.03241 vs. 0.06071), and executing 619 versus 776 trades (20% reduction).

The PETR3-PETR4 pair (Petrobras) exhibited modest CRF improvement from 1.13932 to 1.16663 with Sharpe enhancement from 1.49323 to 1.62987 and reduced maximum drawdown (0.02079 vs. 0.02952), while reducing trades from 482 to 377 (22% reduction). The ITUB3-ITUB4 pair (Itaú Unibanco) achieved 1.39006 versus 1.28718 with dramatically improved Sharpe from 3.12032 to 3.73766 and enhanced Sortino from 5.73131 to 6.20949, executing 689 versus 951 trades (28% reduction).

The BBDC3-BBDC4 pair (Bradesco) represented an exception, with reference method achieving 1.19170 versus Bayesian's 1.18425. The Bayesian approach exhibited higher volatility (0.02109 vs. 0.01787), resulting in lower Sharpe (1.87405 vs. 2.28352) and Sortino (2.96109 vs. 4.40451), though both methods generated substantial absolute returns exceeding 1.15 and the Bayesian method executed 22% fewer trades (419 vs. 534).

5.2.2. Same economic group pairs. Pairs of companies within the same economic group demonstrated predominantly unfavorable performance for the Bayesian approach in this configuration. The CSAN3-RAIZ4 pair showed modest CRF improvement from 1.14176 to 1.16735, though with slightly lower Sharpe (0.66534 vs. 0.70159) due to increased volatility, while significantly reducing trades from 113 to 76 (33% reduction).

Four pairs in this category favored the reference strategy. The ITUB4-ITSA4 (Itaú group) pair demonstrated reference superiority with CRF of 1.20107 versus Bayesian's 1.18405, and substantially higher Sharpe (1.11838 vs. 0.81541) despite the Bayesian method's higher volatility (0.04972 vs. 0.03879), though executing 19% fewer trades (302 vs. 371). The UGPA3-VBBR3 pair showed reference advantage (1.13011 vs. 1.05270) with better Sharpe (0.75628 vs. 0.33686) while the Bayesian method reduced trades from 158 to 115 (27% reduction). The GGBR4-GOAU4 pair (Gerdau group) exhibited slight reference superiority (1.10224 vs. 1.08640) with better Sharpe (0.45666 vs. 0.33698), though the Bayesian approach executed 27% fewer trades (501 vs. 682). The CBAV3-NEOE3 pair demonstrated clear reference advantage (1.04810 vs. 0.98314) with higher Sharpe (0.20757 vs. -0.13342), while the Bayesian method substantially reduced trades from 75 to 40 (47% reduction).

5.2.3. Interpretation of Brazilian market returns. The elevated Sharpe ratios observed for Brazilian dual-class share pairs merit careful interpretation. These results reflect unique structural characteristics of voting versus non-voting share pairs, which are observable across markets. U.S. dual-class pairs such as GOOGL-GOOG (Sharpe: 1.07960),

NWSA-NWS (Sharpe: 1.39911) and FOXA-FOX (Sharpe: 0.60427) demonstrate comparable risk-return profiles, though at lower magnitudes than Brazilian counterparts.

Voting and non-voting shares of the same company bear identical corporate risk, such as exposure to the same management decisions, operational performance, and market conditions. Price divergences therefore reflect transitory mispricing driven by liquidity differentials, voting premium fluctuations, or governance concerns rather than fundamental value differences. This structural similarity, combined with mean-reverting price relationships, explains the characteristically high Sharpe ratios. Convergence trades face minimal exposure to company-specific shocks or economic downturns, as both share classes respond identically to fundamental developments.

The particularly high magnitudes observed in Brazilian markets result from the combination of this structural characteristic with Brazil's less developed market conditions. Emerging markets exhibit greater pricing inefficiencies, wider bid-ask spreads, less algorithmic trading competition, and more persistent deviations from equilibrium relationships compared to developed markets. This creates a more favorable environment for statistical arbitrage strategies, amplifying the base effect present in dual-class pairs globally.

5.3. General remarks

The empirical results demonstrate systematic superiority of the Bayesian approach over the reference cointegration strategy in the majority of analyzed pairs across both U.S. and Brazilian markets, though the magnitude and consistency of improvements differ substantially between these market environments and pair types. In U.S. markets, the Bayesian method outperformed in 18 of 29 equity pairs (62%), while in Brazilian markets it outperformed in 7 of 12 pairs (58%), yielding an overall success rate of 25 of 41 pairs (61%). Although absolute returns in U.S. markets remain modest for many pairs, this performance profile reflects well-documented efficiency constraints in mature equity markets that fundamentally limit statistical arbitrage opportunities. Even within this constrained environment, the Bayesian method demonstrates relative improvement through enhanced signal quality in the majority of cases, as evidenced by superior risk-adjusted performance metrics.

A particularly revealing aspect of our findings emerges from examining the cumulative return factor trajectories in figures 2 and 3. Across both markets and most pair configurations, the reference and Bayesian methods exhibit remarkably similar trading patterns and signal timing, yet with performance gaps that vary in direction and magnitude across different pairs. This pattern indicates that while entry and exit signals occur at approximately the same time points, the Bayesian method's selective signal confirmation produces measurably different results—sometimes superior through filtering marginal entries and premature exits, and occasionally inferior through more conservative trading.

Beyond profitability improvements in successful cases, the Bayesian approach delivers significant operational advantages with direct cost implications. The method executes on average

24% fewer trades than the reference strategy across all analyzed pairs (U.S.: 22.5%; Brazil: 26.6%), reflecting enhanced selectivity in signal generation. This reduction in trading frequency carries substantial practical benefits: proportionally lower transaction costs (broker commissions, exchange fees, market impact), reduced exposure to bid-ask spread costs, and decreased operational complexity.

Risk-adjusted performance metrics favor the Bayesian approach in the majority of analyzed pairs where it achieves superior CRF. Sharpe ratios improve in nearly all such cases, while Sortino ratios demonstrate enhancements in approximately 80% of them, indicating better compensation for risk-taking. Maximum drawdowns are reduced in roughly 60% of successful cases, though not universally. Notably, the Bayesian method tends to operate with higher volatility when it outperforms, accepting increased variance in exchange for superior returns.

The substantial outperformance observed in Brazilian dual-class share pairs likely reflects structural characteristics of emerging equity markets: greater pricing inefficiencies, wider bid-ask spreads, less algorithmic trading competition, and more persistent deviations from equilibrium relationships. The performance pattern observed across both markets indicates variable results across different pair types and market segments. For portfolio implementations, the majority pattern of relative improvement across diverse market conditions suggests potential for enhanced risk-adjusted returns through diversification, though careful pair selection based on structural characteristics remains essential for optimal deployment.

6. Simulation study for pair selection

In this section, we evaluate the proposed method's effectiveness in identifying false positives from cointegration tests through simulation studies. While cointegration tests can yield both false positives and false negatives, false positives present a greater risk to investors. Trading pairs incorrectly identified as cointegrated can lead to substantial losses, as their prices may not maintain the assumed long-term equilibrium relationship. This risk amplifies when conducting multiple tests. Our proposed Bayesian method addresses this challenge by enhancing the pair selection process, offering a heuristic mechanism to detect potential false positives.

The cointegration test methodology includes a unit root test on the residuals as its second step, where the null hypothesis assumes these residuals are non-stationary. When evaluating this assumption, a type I error occurs if we incorrectly reject a true null hypothesis, leading to a false identification of cointegration. In other words, we might conclude that two price series are cointegrated when they actually have no long-term equilibrium relationship. Although we can theoretically control the type I error rate by establishing a significance level α , there is a specific situation in which we can significantly inflate the probability of observing false positives when applying the cointegration test in pairs trading. Clegg (2014) states that the use of a data mining procedure to identify cointegrated pairs on a list of potential candidates tends to inflate the chances of false positives.

If we perform multiple cointegration tests sequentially to identify potentially cointegrated pairs in an exchange, the probability of observing at least one false positive increases beyond the nominal significance level α . This accumulation of Type I errors is known as the family-wise error rate (FWER) problem. Selecting non-cointegrated pairs is particularly risky because the assumed long-term equilibrium is spurious, resulting in unreliable trading signals.

In this context, correction methods based on the False Discovery Rate (FDR), such as the Benjamini-Hochberg procedure (Benjamini and Hochberg 1995), are unsuitable because they control only the expected proportion of false positives among rejections, still allowing erroneous selections that can incur significant costs. Conversely, methods that control the FWER, such as the Bonferroni correction (Bonferroni 1936, Haynes 2013) or the Holm-Bonferroni procedure (Holm 1979), are overly conservative. While effective at reducing false positives, they substantially inflate the Type II error rate, rejecting many genuinely cointegrated pairs. This limitation is particularly problematic in data-mining approaches involving a large number of tests. For instance, considering the S&P 500 index, the number of possible pairs exceeds 500². In such scenarios, the Type II error rate becomes so severe that nearly all pairs would be rejected, potentially excluding viable investment opportunities.

With that in mind and acknowledging the benefits of using the Bayesian framework, we hypothesized in Section 3.1 that the proposed method is not only capable of generating more accurate trading signals but also executing a more precise pair selection. This is mainly due to the adoption of a truncated normal as the prior distribution of β , which also causes the full conditional to be a truncated normal. As presented in Section 3.1, by truncating the distribution at $\beta_1 < 0$, we can identify pairs that do not move together most of the time—a basic premise of the pairs trading strategy. This adds an extra layer of confirmation to the cointegration test, and, as we hypothesize, can help reduce the number of false positives through prior beliefs and heuristics.

The empirical results have shown that the Bayesian method delivers a superior performance in terms of profitability and risk management when compared to standard cointegration. Nonetheless, in order to evaluate the specific hypothesis of enhancement of the pair selection, we ran a set of simulations with both cointegrated and non-cointegrated series to determine if our algorithm is capable of detecting false positives in the cointegration test.

6.1. Design and results

A total of 60,000 simulations were performed, organized into six batches of 10,000 simulations each. For each batch, 1,000 samples were generated from each of six selected sample sizes {60, 90, 120, 180, 252, 500}, and combined with five different β_1 values {0.5, 0.75, 1.0, 1.25, 1.5} for the ordinary least squares (OLS) regression between the series y_t and x_t . Lastly, each possible combination of data size and β_1 was simulated in two scenarios—one in which the two series are co-integrated and one in which they are not.

Table 3. Performance of the Bayesian method in detecting false positives from cointegration tests across different sample sizes and β_1 values.

	Sample Size						Total
	500	252	180	120	90	60	
$\beta_1 = 0.50$							
Cointegration False Positives	73	36	70	71	52	0	302
Discarded by Bayesian Layer	43	17	70	70	35	0	235
Proportion	0.59	0.47	1.00	0.99	0.67	–	0.78
$\beta_1 = 0.75$							
Cointegration False Positives	73	36	70	72	34	2	287
Discarded by Bayesian Layer	44	19	70	72	17	1	223
Proportion	0.60	0.53	1.00	1.00	0.50	0.50	0.78
$\beta_1 = 1.00$							
Cointegration False Positives	76	33	70	72	68	0	319
Discarded by Bayesian Layer	43	16	70	72	17	0	218
Proportion	0.57	0.48	1.00	1.00	0.25	–	0.68
$\beta_1 = 1.25$							
Cointegration False Positives	71	37	70	71	37	1	287
Discarded by Bayesian Layer	41	20	70	70	18	0	219
Proportion	0.58	0.54	1.00	0.99	0.49	0.00	0.76
$\beta_1 = 1.50$							
Cointegration False Positives	75	35	72	71	17	0	270
Discarded by Bayesian Layer	45	17	72	70	0	0	204
Proportion	0.60	0.49	1.00	0.99	0.00	–	0.76
Overall Totals							
Cointegration False Positives	368	177	352	357	208	3	1465
Discarded by Bayesian Layer	216	89	352	354	87	1	1099
Proportion	0.587	0.503	1.00	0.99	0.418	0.33	0.75

Note: The table reports the number of false positives from the cointegration test, the number discarded by the Bayesian layer, and the corresponding detection rate for each combination of sample size and β_1 .

To simulate a cointegrated relationship, we generate two series, $y_t = \sum_{i=1}^t \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, 1)$ i.i.d., and $x_t = \beta_1 y_t + \eta_t$, where $\eta_t \sim \mathcal{N}(0, 1)$ i.i.d. The two series are non-stationary because they are cumulative sums of white noise, i.e. they follow a random walk. However, since x_t is a linear combination of y_t with an additional noise component, the regression of y_t on x_t should produce stationary residuals, thus characterizing the cointegration.

For a non-cointegrated relationship, we start by also simulating two non-stationary series, $y_t = \sum_{i=1}^t \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, 1)$ i.i.d., and $x_t = \sum_{i=1}^t v_i$, where $v_i \sim \mathcal{N}(0, 1)$ i.i.d. Nonetheless, x_t is not represented as a linear combination of y_t anymore. Thus, regressing one on the other should not reveal a significant long-term relationship since both series are generated independently (each as a separate random walk). Therefore the residuals from this regression should be non-stationary.

Once we have the original series and their true nature, each simulated run is tested for cointegration using the Phillips-Perron framework at $\alpha = 0.05$ significance level and the result is stored. Our method is applied immediately by drawing realizations from the full conditional of β . If a sequence of $m = 30$ negative β_1 values is drawn, then an annotation indicating the pair should be discarded is stored in our database.

When there was a false positive result from the cointegration test and our method correctly suggested discarding the pair we counted this as a correct decision. We also examined potential errors in the proposed method related to erroneously discarding pairs. A discarding error was recorded whenever

our method recommended rejecting a pair that was, in fact, cointegrated, regardless of the cointegration test outcome.

Table 3 below show the summary of the results obtained in the simulations. Our method correctly identified around 75% of the total false positives in the simulations. Although it is not shown in the table, it is also important to highlight that the algorithm did not erroneously discard any truly cointegrated pairs.

In summary, the simulation results indicate that the Bayesian method can correctly identify a considerable amount of false positive results from cointegration tests across various data sizes and β_1 values without erroneously discarding truly cointegrated pairs. The results suggest that this approach can serve as a valuable tool for enhancing cointegration analysis in general applications and not only in pairs trading. Further research and refinement may improve the method's precision in pair selection and extend its application to other contexts. Future work could also focus on deriving theoretical results that would allow a transition from its current heuristic state to a more theoretically grounded approach.

7. Concluding remarks

We developed a Bayesian method for pairs trading that enhances traditional cointegration strategies by dynamically adjusting trading signals based on the full conditional distribution of the hedge ratio β_1 . The methodology incorporates Bayesian hierarchical modeling with truncated normal priors

to improve both pair selection and signal generation compared to the standard cointegration strategy.

The empirical validation across U.S. and Brazilian markets confirms the method's effectiveness and adaptability to different market structures. The results indicate systematic improvements in profitability and risk-adjusted performance, with the Bayesian method outperforming in 62% of U.S. pairs (18 of 29) and 58% of Brazilian pairs (7 of 12). While the method does not universally dominate across all pair configurations, the consistent pattern of relative advantage in the majority of cases demonstrates the value of distribution-based signal confirmation. The method demonstrates particularly strong results in dual-class share pairs. Notably, performance gains were considerably higher in magnitude in emerging markets. Brazilian pairs exhibited higher absolute returns and Sharpe ratios compared to developed market counterparts, reflecting the effectiveness of the method in environments with greater pricing inefficiencies and structural opportunities.

Beyond profitability improvements, the Bayesian framework delivers significant operational advantages. The method's selectivity in signal generation reduces trading frequency by approximately 24% on average (22.5% in U.S. markets, 26.6% in Brazilian markets), translating to proportionally lower transaction costs and reduced market impact in practical implementations. This efficiency gain represents a crucial advantage for institutional deployment, where transaction costs can significantly erode strategy profitability.

The methodology's scope extends beyond pairs trading applications. Our simulation studies reveal that the Bayesian approach considerably filters out false positives in cointegration testing, offering a more refined statistical tool for cointegration detection. This finding suggests broader applicability in econometric analysis beyond quantitative finance.

Several research directions warrant further investigation. Exploring alternative methods for deriving the conditional distribution of the hedge ratio β_1 represents a promising avenue, as different distributional assumptions or computational approaches may enhance the framework's adaptability and performance across diverse market conditions. Incorporating realistic transaction cost models would provide more precise performance assessments and quantify the full operational advantage of reduced trading frequency. Extending the framework to additional asset classes, developing adaptive parameter selection mechanisms, and formalizing the theoretical foundations of our pair selection approach represent additional promising avenues for advancing the methodology's practical and theoretical contributions.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendix

Table A1. Top 3 correlated stock pairs by sector (correlation ≥ 0.8).

Sector	Stock pair	Correlation
Communication services	GOOGL – GOOG	0.9957
	FOXA – FOX	0.9876
	NWSA – NWS	0.9780
Consumer discretionary	DHI – LEN	0.9184
	DHI – PHM	0.8955
	HLT – MAR	0.8954
Consumer staples	N/A	N/A
Energy	MPC – VLO	0.8749
	COP – EOG	0.8731
	HAL – SLB	0.8604
Financials	CFG – FITB	0.8885
	MET – PRU	0.8879
	FITB – HBAN	0.8867
Healthcare	N/A	N/A
Industrials	RSG – WM	0.8079
	ADP – PAYX	0.8021
Information technology	AMAT – LRCX*	0.9266
	AMAT – KLAC	0.9110
	KLAC – LRCX	0.9107
Materials	MLM – VMC	0.9087
	DOW – LYB	0.8826
	NUE – STLD	0.8820
Real estate	EQR – UDR	0.9287
	FRT – REG	0.9262
	AVB – EQR	0.9241
Utilities	CMS – WEC	0.8985
	CMS – DTE	0.8867
	LNT – WEC	0.8857

Note: Data and period: daily adjusted close prices from 02/01/2015 to 01/31/2025 (10 years)

Total pairs analyzed: 168 pairs with correlation ≥ 0.8

*In the analysis, the pair AMAT-LCRX was replaced by KLAC-LRCX due to inconsistencies found in the price data. No pairs with daily correlation ≥ 0.8 were found for the Consumer Staples and Healthcare sectors.

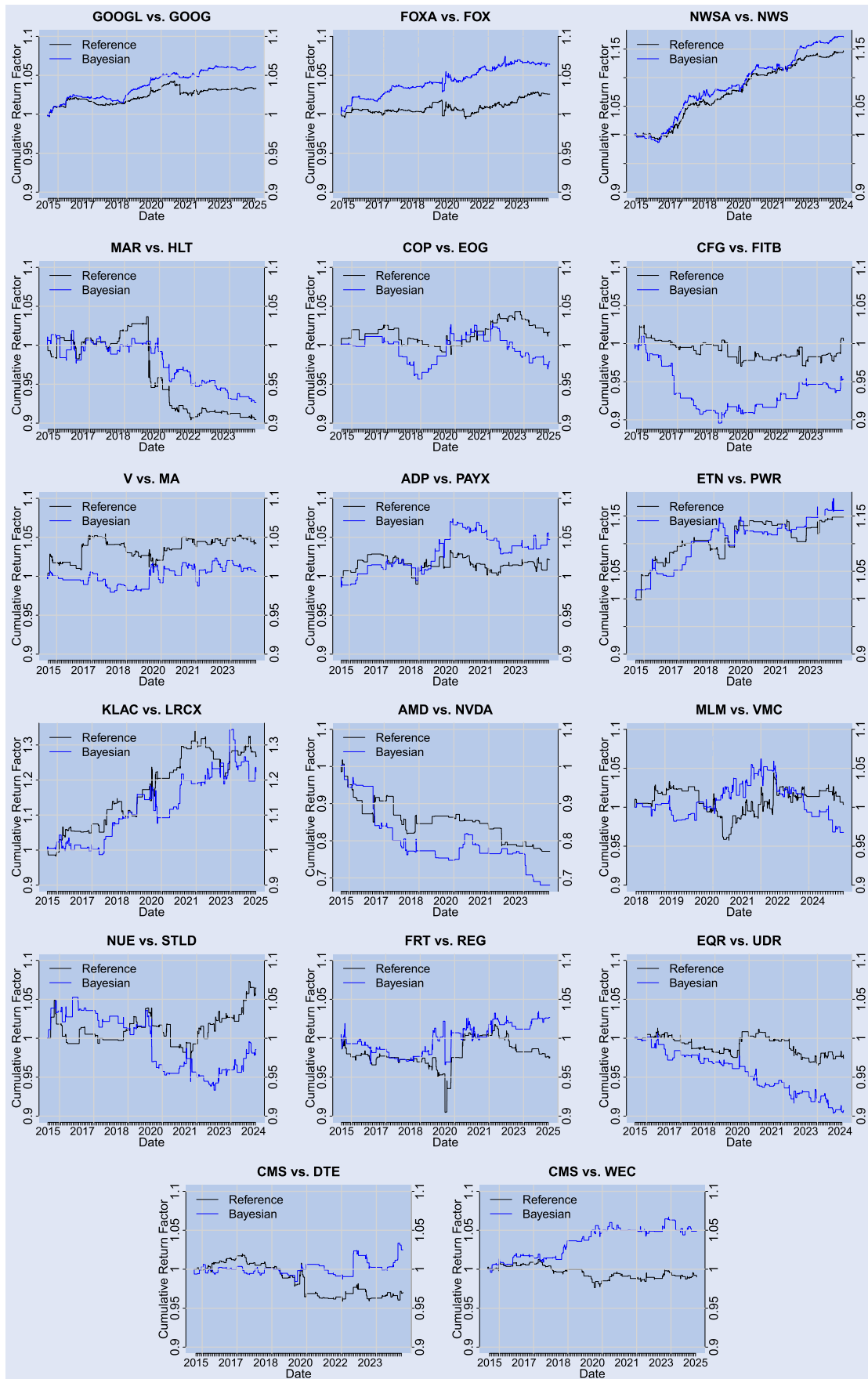


Figure A1. Cumulative return factor comparison for additional US equity pairs not shown in Figure 2 (02/01/2015–01/31/2025). Blue curves represent the Bayesian method; black curves represent the reference cointegration strategy.